



Explainable AI for Credit Risk Assessment

Explaining the Black Box: Comparing Explanation
Methods, Frameworks and Fairness in Credit Risk
Modelling

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Abstract

Post hoc explanation methods are increasingly used to interpret automated credit risk models, yet the reliability of the explanations they produce cannot be assumed. This study evaluates three widely used feature attribution methods, SHAP, LIME and Integrated Gradients, across four classifiers: logistic regression, XGBoost, and matched feed forward neural networks implemented in TensorFlow and PyTorch. Two credit datasets with substantially different sample sizes are examined: the Taiwan Default of Credit Card Clients dataset, containing 30,000 observations, and the German Statlog Credit dataset, containing 1,000 observations.

Three aspects of explanation reliability are investigated. First, agreement between explanation methods is assessed using Kendall’s coefficient of concordance and pairwise Spearman rank correlations. Second, reproducibility across the TensorFlow and PyTorch implementations is evaluated using paired Wilcoxon signed rank tests across ten matched random seeds with Bonferroni correction. Third, the relationship between demographic feature attribution and group level outcome disparity is examined using the four fifths screening criterion with bootstrap confidence intervals.

Overall concordance between SHAP, LIME and Integrated Gradients was statistically significant for all neural network analyses, with Kendall’s W ranging from 0.605 to 0.944. Pairwise results nevertheless revealed substantial variation between explanation methods. SHAP and Integrated Gradients showed consistently strong agreement, with Spearman correlations ranging from 0.890 to 0.952. Comparisons involving LIME were markedly weaker for the Taiwan neural networks, whereas substantially stronger agreement was observed on the German dataset. These results demonstrate that statistically significant overall concordance does not imply that every explanation method produces closely matching feature rankings.

The framework comparison showed strong agreement in feature ordering between TensorFlow and PyTorch, with framework level Spearman correlations of 0.957 for Taiwan and 0.962 for German. No Taiwan features showed statistically detectable differences after Bonferroni correction, whereas two of the 20 German features did. Attribution magnitudes were nevertheless

not identical, particularly on the smaller German dataset, meaning that high ranking reproducibility should not be interpreted as formal equivalence between the two implementations.

Age produced the clearest demographic outcome disparity. For Taiwan logistic regression, the age four fifths ratio was 0.680 with a bootstrap confidence interval of 0.490 to 0.762. On the German dataset, all four models produced age ratios below 0.8, with their bootstrap confidence intervals also remaining below the screening threshold, although these estimates were based on a substantially smaller test sample. Feature attribution results provided complementary rather than equivalent evidence, demonstrating that low direct attribution to a demographic feature does not preclude group level disparity.

Taken together, the findings show that explanation reliability is conditional on the modelling and data setting. Differences in data availability and the behaviour of the fitted models provide a plausible context for several of the cross dataset patterns, although the two-dataset design does not establish a causal or universal relationship. The principal contribution of the study is therefore a reproducible comparative framework for evaluating explanation method agreement, implementation reproducibility and demographic outcome disparity rather than assuming that post hoc explanations are inherently stable or interchangeable.

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Glossary and Abbreviations

Abbreviations

Abbreviation	Expansion
AUC	Area under the receiver operating characteristic curve
CI	Confidence interval
EEOC	Equal Employment Opportunity Commission (United States)
GDPR	General Data Protection Regulation
IG	Integrated Gradients
LIME	Local Interpretable Model Agnostic Explanations
NN	Neural network
ReLU	Rectified Linear Unit
ROC	Receiver operating characteristic
RQ1, RQ2, RQ3	Research Questions One, Two and Three
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
UCI	University of California, Irvine Machine Learning Repository
XAI	Explainable Artificial Intelligence

Terms

Term	Definition as used in this dissertation
Attribution	A numerical value assigned by an explanation method to a feature, quantifying that feature's contribution to a model prediction.
Bonferroni correction	An adjustment used to control the family wise error rate when multiple hypotheses are tested, applied in this study to the feature level framework comparisons.

Term	Definition as used in this dissertation
Bootstrap	A resampling procedure used to estimate sampling variability, applied in this study to construct confidence intervals for the four fifths ratios.
Concordance	The degree to which two or more explanation methods produce similar rankings of features by importance.
Demographic attribute	A characteristic used in the demographic analysis, such as age or sex.
Disparate impact	A difference in favourable outcome rates between demographic groups, considered here as a group level screening measure rather than a determination of unlawful discrimination.
Four fifths rule	A screening criterion under which a favourable outcome ratio below 0.8 is treated as a signal requiring further examination.
Friedman test	A nonparametric test for differences between three or more related samples, used here to compare model performance across common cross validation folds.
Implementation invariance	The attribution axiom stating that functionally equivalent networks should receive identical attributions even if their internal implementations differ.
Information Value	A univariate measure of association between a feature and a binary outcome, calculated using weight of evidence binning and commonly used in credit scorecard development.
Kendall's coefficient of concordance (W)	A statistic ranging from 0 to 1 that measures agreement among multiple rankings of the same items.
Nemenyi test	A post hoc procedure used following a statistically significant Friedman test to identify differences between pairs of models.
One hot encoding	The representation of a categorical variable using a set of binary indicator variables.

Term	Definition as used in this dissertation
Post hoc explanation	An explanation of model behaviour generated after the predictive model has been trained, without changing the fitted model itself.
Shapley value	A concept from cooperative game theory that allocates a payoff among contributors according to their marginal contributions and forms the theoretical basis of SHAP.
Spearman rank correlation	A measure of monotonic association between two rankings, used in this study to examine pairwise agreement between explanation methods and framework level feature rankings.
Stratified split	A division of observations into subsets while approximately preserving the class proportions of the original data.
Wilcoxon signed rank test	A nonparametric test for paired observations, used in this study with matched random seeds providing the pairing for the TensorFlow and PyTorch comparisons.

Chapter 1

Introduction

Credit scoring plays a crucial role in determining who receives a loan, mortgage, or credit card, as well as the terms of those financial products. For decades, the industry relied on logistic regression scorecards, which provided relatively traceable decisions based on a limited number of interpretable factors. Machine learning approaches have changed this landscape, with gradient boosted ensembles and neural networks showing stronger predictive performance than traditional scorecards in some settings (Lessmann et al. 2015). However, more complex models can also be more difficult to interpret because their internal decision making processes are not directly accessible in the same way as those of conventional scorecards (Barredo Arrieta et al. 2020).

This lack of transparency matters because credit decisions affect identifiable individuals and can produce uneven outcomes across demographic groups (Fuster et al. 2022). An applicant denied credit by an opaque model may reasonably require an explanation of the factors that contributed to that decision, while auditors and regulators require reliable accounts of model behaviour. A misleading explanation is therefore problematic because it can create confidence in an account that does not accurately represent the underlying model.

The issue also has regulatory significance. Data protection requirements relating to automated decision making include meaningful information about the logic involved and safeguards for affected individuals (European Union 2016). In the United Kingdom, the Financial Conduct Authority’s Consumer Duty places emphasis on good outcomes and consumer understanding (Financial Conduct Authority 2022). At European Union level, the Artificial Intelligence Act generally classifies AI systems used to evaluate creditworthiness or establish credit scores as high risk applications, subject to its scope and exceptions (European Union 2024). Post hoc explanation methods such as SHAP (Lundberg and Lee 2017), LIME (Ribeiro, Singh, and Guestrin 2016) and Integrated Gradients (Sundararajan, Taly, and Yan

2017) provide ways of interpreting complex predictive models.

However, generating an explanation does not establish that it is reliable. Different explanation methods may produce different feature rankings, independently trained implementations may not produce identical attribution patterns, and apparently reassuring feature attributions do not establish that model outcomes are equitable across demographic groups. These concerns motivate the three research questions examined in this study.

Table 1.1 summarises the three attribution methods and the questions motivating their comparison.

1.1 Research Questions

Three questions therefore follow, each evaluated using a prespecified statistical analysis rather than visual inspection alone.

- RQ1.** Do SHAP, LIME and Integrated Gradients agree on the global ranking of features important to model predictions, as measured using Kendall's coefficient of concordance?
- RQ2.** Does implementing a matched neural network architecture in TensorFlow and PyTorch produce differences in feature attribution or training behaviour across multiple random seeds?
- RQ3.** Do the models exhibit demographic disparities under the four fifths screening criterion, and how do the resulting group level disparities correspond with the importance assigned to demographic attributes by the explanation methods?

1.2 Study Design

To address these questions, four models are trained on each of two credit datasets: logistic regression, XGBoost, and matched neural networks implemented in TensorFlow and PyTorch. The Taiwan Default of Credit Card Clients dataset and the German Statlog Credit dataset provide a direct comparison under substantially different data conditions.

The analysis compares explanation agreement, framework related variation, pre-

Table 1.1: Strengths, limitations and open questions for the three attribution methods evaluated in this study.

Method	Basis	Strengths	Limitations	Open question addressed here
SHAP (Lundberg and Lee 2017)	Shapley values from cooperative game theory	Local accuracy and consistency; exact TreeSHAP available for tree ensembles (Lundberg, Erion, et al. 2020)	Computationally expensive; game theoretic framing may not match the practical question (Kumar et al. 2020)	Whether its rankings agree with other methods
LIME (Ribeiro, Singh, and Guestrin 2016)	Local linear surrogate fitted to sampled perturbations	Model agnostic; applicable to any classifier	Sensitive to perturbation sampling (Ghorbani, Abid, and Zou 2019) and open to manipulation (Slack et al. 2020)	Whether its agreement with other methods varies across settings
Integrated Gradients (Sundararajan, Taly, and Yan 2017)	Path integration of model gradients from a baseline	Satisfies sensitivity and implementation invariance	Restricted to differentiable models; sensitive to baseline choice	Whether rankings remain consistent across matched neural network implementations

dictive performance and demographic fairness across both datasets. This provides evidence beyond a single modelling setting, but two datasets cannot establish that the same patterns will occur across all credit datasets or application domains. Generalisation beyond the evaluated datasets is therefore treated as an empirical question. The reproducible analysis pipeline allows the same procedures to be applied to additional datasets.

The results are interpreted together to examine whether differences in data availability and fitted model behaviour may help explain variation in attribution agreement, framework sensitivity and demographic outcomes. These factors are treated as possible interpretations of cross dataset patterns rather than as causal assumptions.

1.3 Contributions

The first contribution is a comparative evaluation of explanation method agreement across models and datasets. SHAP, LIME and Integrated Gradients are compared using common feature representations and formal rank based measures, allowing agreement and disagreement to be quantified rather than inferred from visual inspection.

The second contribution is an empirical comparison of matched neural network implementations in TensorFlow and PyTorch. The networks use matched architectures, data inputs and training settings and are evaluated across ten random seeds. This tests the extent to which independently trained implementations produce similar predictive behaviour and feature attribution patterns while recognising that implementation and execution differences remain.

The third contribution connects feature attribution analysis with a formal group level fairness screening measure. Demographic feature importance is compared with favourable outcome ratios calculated using the four fifths criterion, allowing the two forms of evidence to be examined as complementary rather than interchangeable.

Finally, the two-dataset design examines whether findings recur under contrasting data conditions. It does not establish universal generalisability, but provides a comparative procedure that can be applied to additional datasets.

The originality of the project lies in bringing these dimensions together within a single reproducible comparative design, allowing explanation method agreement,

implementation reproducibility and demographic outcome disparity to be evaluated as related but distinct aspects of explanation reliability.

1.4 Project Plan

The project was carried out between June and September 2026. Initial work focused on developing an understanding of explainable artificial intelligence in credit risk modelling, completing the required ethics documentation and refining the research design. Following the first supervisor meeting, the proposed data source was reconsidered and the Taiwan and German UCI datasets were selected. Data preparation was then followed by development of the predictive models and implementation of SHAP, LIME and Integrated Gradients.

The main analytical work was completed during July. Statistical analysis was undertaken alongside the three research questions, examining explanation method agreement, reproducibility between the TensorFlow and PyTorch implementations, and demographic fairness. Dissertation writing progressed alongside the analytical work, with the methodology and results developed during July and the remaining chapters prepared for the first full draft in early August.

Feedback from supervisor meetings in June, July and August informed subsequent revisions. Later work included conversion to the Loughborough University template, refinement of figures and equations, organisation of the code and GitHub repository, and final proofreading and formatting. Following submission of the dissertation, the remaining project period was allocated to preparation for the PowerPoint submission and oral presentation.

Figure 1.1 provides a visual overview of the project timeline, showing how the main research, modelling, analysis, writing and revision stages progressed alongside the confirmed project milestones. Concise activity labels are used within the chart for readability, with the principal stages described in greater detail in the preceding paragraphs.

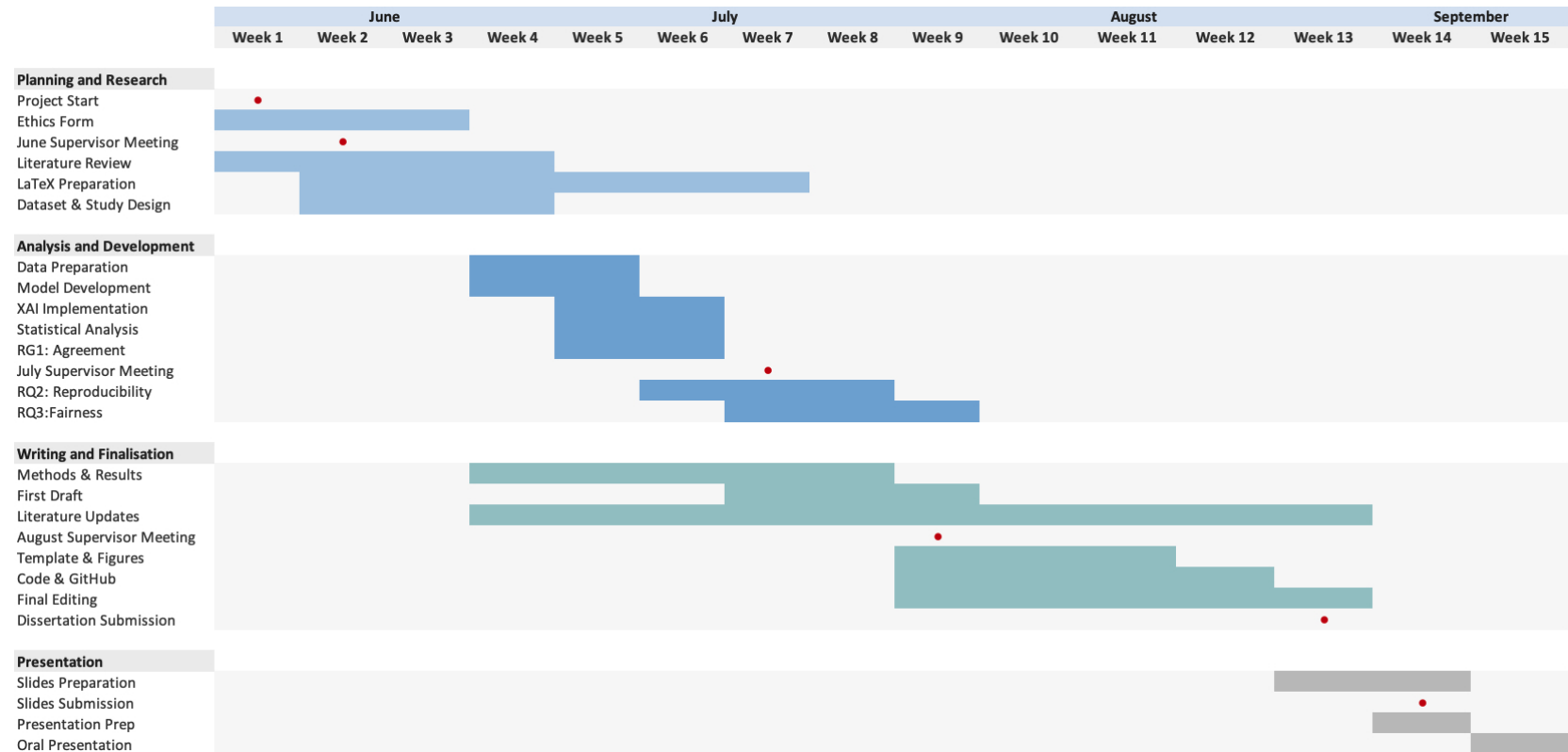


Figure 1.1: Dissertation project Gantt chart. Red circles mark confirmed milestones; coloured bars show approximate activity periods.

1.5 Dissertation Structure

Chapter 2 reviews the literature on explainable artificial intelligence, credit risk modelling, explanation agreement, implementation consistency and fairness. Chapter 3 describes the datasets, preprocessing, models, explanation methods and statistical methodology. Chapter 4 presents the empirical results, Chapter 5 discusses them in relation to the research questions and literature, and Chapter 6 concludes the dissertation. Supporting material is provided in the Appendix.

Chapter 2

Literature Review

2.1 Introduction

This study builds on three established research areas: machine learning for credit scoring, post hoc explanation methods, and algorithmic fairness. The credit scoring literature demonstrates that flexible models can improve predictive performance but may create additional interpretability challenges. Explainability research shows that explanations can disagree, vary in stability and be manipulated, while fairness research provides formal measures for comparing outcomes across demographic groups.

The questions in this study arise where these areas overlap. The review first considers the interpretability problem in credit scoring, before examining the theoretical foundations and reliability of the explanation methods. It then considers reproducibility across implementations and the relationship between demographic outcome disparities and feature attribution. These areas provide the basis for the three research questions.

2.2 Machine Learning in Credit Scoring and the Interpretability Tension

Logistic regression scorecards have traditionally played an important role in credit scoring because they combine predictive capability with a relatively transparent representation of the factors contributing to an applicant’s score (Siddiqi 2006). The increasing use of machine learning approaches has challenged this position by demonstrating that more flexible models can achieve stronger predictive performance.

Evidence for this performance advantage is well established, although it is not

uniform across model types. Lessmann et al. (2015) compared a broad range of classification methods and demonstrated the competitive performance of ensemble and other machine learning approaches relative to conventional statistical models. The systematic review by Dastile, Celik, and Potsane (2020) similarly found that ensemble approaches frequently perform strongly in credit scoring applications, while identifying explainability and class imbalance as recurring challenges. These findings suggest that improved predictive capability may come with additional practical considerations, particularly where model decisions need to be understood or audited.

Greater model complexity does not, however, guarantee better performance. Gunnarsson et al. (2021) found that deeper neural networks do not consistently outperform gradient boosted approaches in credit scoring. This challenges the assumption that increasingly complex architectures are necessarily preferable and supports the inclusion of XGBoost alongside relatively modest neural network architectures in the present study.

Predictive improvements may also be distributed unevenly. Fuster et al. (2022) found that machine learning improvements in mortgage lending can differ between demographic groups. Flexible models may also exploit relationships among variables that act as proxies for demographic characteristics. Consequently, improved overall predictive performance does not establish that model behaviour is equally beneficial or equally interpretable across groups. The move towards more complex models therefore raises both interpretability and fairness concerns.

Credit scoring data commonly contain class imbalance because defaulting applicants represent a minority of observations. Synthetic oversampling, including SMOTE (Chawla et al. 2002), is one possible response, while class weighting provides an alternative that does not create synthetic applicant records. This is particularly useful for an explainability study because the observations being explained remain associated with genuine records.

Accuracy alone can also be misleading for imbalanced classification. ROC AUC is therefore widely used in credit scoring model comparisons, including Lessmann et al. (2015), and is used as the principal comparative performance measure in this study.

Taken together, this literature establishes the central tension considered here:

predictive performance alone is insufficient where the resulting model must also be interpreted and audited. However, demonstrating that explanations are necessary does not establish that a particular explanation is reliable. The explanation methods themselves must therefore be examined.

2.3 Post hoc Explanation Methods and Their Foundations

SHAP, LIME and Integrated Gradients represent three widely used approaches to post hoc feature attribution, but they are based on different theoretical foundations.

SHAP was introduced by Lundberg and Lee (2017) and derives feature attributions from Shapley values in cooperative game theory. Within the additive feature attribution framework, SHAP satisfies properties including local accuracy, missingness and consistency. Model specific implementations have made it computationally practical for particular model classes, including TreeSHAP for tree ensembles (Lundberg, Erion, et al. 2020).

Integrated Gradients was proposed by Sundararajan, Taly, and Yan (2017). It attributes model output by integrating gradients along a path between a reference baseline and the observation being explained. The method was designed to satisfy sensitivity and implementation invariance. Implementation invariance states that functionally equivalent networks should receive identical attributions, but it does not imply that independently trained neural networks will learn the same function.

LIME takes a different approach. Ribeiro, Singh, and Guestrin (2016) proposed Local Interpretable Model Agnostic Explanations as a method that approximates model behaviour around an individual observation using a locally fitted interpretable surrogate. Unlike Integrated Gradients, it does not require access to model gradients and can therefore be applied across different classifier types.

These methods occupy different positions within the taxonomy of explainable artificial intelligence described by Barredo Arrieta et al. (2020). LIME is explicitly model agnostic, SHAP can use model agnostic or model specific explainers, and Integrated Gradients is restricted to differentiable models. Their different assumptions and model requirements mean that similar outputs should not be assumed simply because each method produces feature level attributions.

Gradient based attribution also has a longer history. Saliency methods use the gradient of model output with respect to the input as a measure of local importance (Simonyan, Vedaldi, and Zisserman 2014), while DeepLIFT compares activations against a reference (Shrikumar, Greenside, and Kundaje 2017). Integrated Gradients similarly uses a reference point but integrates gradients along the path to the observation (Sundararajan, Taly, and Yan 2017).

This shared use of gradients is relevant because the neural network SHAP implementation used in this study is also gradient based. SHAP and Integrated Gradients therefore provide related, although not identical, forms of attribution, whereas LIME relies on perturbation sampling and a local surrogate. Their different foundations provide a theoretical reason for both agreement and disagreement, making empirical comparison more informative than assuming that the methods are interchangeable.

Theoretical properties also do not guarantee that an explanation is appropriate for every practical question. Kumar et al. (2020), for example, argue that Shapley value explanations can be misleading where the underlying game theoretic formulation does not correspond to the question being asked. Explanation quality therefore depends not only on satisfying formal properties but also on how the explanation is used and interpreted. Agreement between explanation methods consequently remains an empirical issue.

2.4 Beyond Feature Attribution: Counterfactuals and Recourse

Feature attribution is only one approach to explainable artificial intelligence. Counterfactual explanations instead identify changes to an individual’s characteristics that would result in a different prediction. Wachter, Mittelstadt, and Russell (2017) developed this approach as a way of explaining automated decisions without requiring disclosure of the complete internal model.

Anchors provides a related approach based on conditions that are sufficient to support a prediction within a defined neighbourhood (Ribeiro, Singh, and Guestrin 2018). Ustun, Spangher, and Liu (2019) further developed actionable recourse by distinguishing between characteristics that an individual can realistically change

and those that cannot. Counterfactual approaches have also been applied directly to credit scoring problems (Dastile, Celik, and Vandierendonck 2022).

These methods answer a different question from feature attribution. Counterfactual and recourse methods ask how an outcome might be changed, whereas feature attribution identifies variables associated with a model’s output. The present study requires a common representation of feature importance to compare explanation methods, frameworks and demographic attributes. Feature attribution is therefore the focus, without implying that it provides a complete account of explainability.

2.5 The Reliability of Explanations: Agreement, Stability and Manipulation

Generating an explanation does not guarantee that it is trustworthy. One concern is disagreement between explanation methods. Krishna et al. (2022) formalised this as the disagreement problem and showed that widely used methods can produce substantially different feature rankings across models and datasets. Evidence from credit scoring is more mixed, with Gramegna and Giudici (2021) reporting meaningful agreement between SHAP and LIME in a credit risk setting. Taken together, these findings suggest that concordance is unlikely to be a fixed property of the explanation methods and may instead depend on the model, dataset and explanation setting.

Stability is a second concern. Ghorbani, Abid, and Zou (2019) demonstrated that feature attributions can change substantially following small input perturbations even where the model prediction remains largely unchanged. This creates a distinction between predictive stability and explanation stability: a model may behave consistently while the explanation of that behaviour changes. An explanation should therefore not be treated as reliable solely because the underlying prediction is stable.

Manipulability presents a further problem. Slack et al. (2020) showed that post hoc explanation methods including LIME and SHAP can be deceived by specially constructed models, producing apparently acceptable explanations despite discriminatory underlying behaviour. This demonstrates that an explanation can appear plausible without providing a faithful account of the behaviour that is important

for auditing.

These concerns have encouraged more quantitative evaluation of explanation quality. Alvarez-Melis and Jaakkola (2018) examined robustness as the extent to which similar inputs receive similar explanations, while OpenXAI (Agarwal et al. 2022) provides a framework for reproducible evaluation of explanation methods. Although these approaches evaluate different aspects of explanation quality, they support the wider principle that reliability should be measured rather than assumed.

The concordance analysis in this study follows this principle. Kendall’s coefficient of concordance provides an overall measure of agreement, while pairwise Spearman correlations identify where disagreement is concentrated. Agreement is therefore evaluated explicitly rather than inferred from visually similar explanation outputs.

Existing research establishes that explanation methods can disagree, be unstable and be manipulated, but the degree to which agreement changes across modelling and data conditions remains less clear. The two-dataset design used here provides comparative evidence on this question, although it cannot isolate dataset size or model behaviour as causal factors because the datasets differ in several respects. This limitation is important when interpreting any cross dataset differences observed in Research Question 1.

2.6 The Reproducibility of Explanations Across Implementations

A second reliability question is whether explanation results remain similar across independently trained implementations. This is distinct from disagreement between explanation methods because variation can arise even when the same modelling approach and attribution procedure are intended to be reproduced.

Implementation invariance, introduced by Sundararajan, Taly, and Yan (2017), states that two models computing the same mathematical function should produce identical Integrated Gradients attributions. It does not state that independently trained TensorFlow and PyTorch networks will converge to the same function. Differences in initialisation, stochastic training and numerical implementation may therefore produce networks with similar predictive performance but different learned relationships.

This distinction matters where explanations are used in model auditing. Bückner et al. (2022) argue for systematic evaluation of explainability within model risk management, where stability and reproducibility are important characteristics. If closely matched implementations produce materially different explanations, the resulting attribution may be more difficult to interpret as a stable property of the modelling process.

Machine learning results may vary because of random initialisation, software versions, hardware and other execution details. Explanation outputs can therefore inherit variation from both model training and the attribution procedure. Similar predictive accuracy does not establish that two trained networks have learned identical functions or feature relationships, meaning that predictive reproducibility and explanation reproducibility should be considered separately.

The literature does not fully establish how large this variation is under closely matched cross framework conditions. Research Question 2 addresses this by comparing matched TensorFlow and PyTorch networks across ten random seeds using common data inputs and closely aligned training settings. The analysis measures the variation observed between the implementations rather than claiming to isolate a causal framework effect.

2.7 Fairness, the Four Fifths Rule and the Limits of Fairness Metrics

Algorithmic fairness includes several different definitions. Surveys such as Mehrabi et al. (2021) and Barocas, Hardt, and Narayanan (2023) distinguish between group and individual criteria. Demographic parity concerns differences in outcome rates between groups, while equalised odds and equality of opportunity concern differences in error rates conditional on the true outcome (Hardt, Price, and Srebro 2016). Fairness through awareness instead focuses on similar treatment for similar individuals (Dwork et al. 2012). These criteria answer different questions and cannot be treated as interchangeable, meaning that the choice of fairness measure determines which aspect of model behaviour is being assessed.

The four fifths rule originates in the United States Uniform Guidelines on Employee Selection Procedures (Equal Employment Opportunity Commission 1978). It is

therefore an employment selection criterion rather than a lending specific legal test. Feldman et al. (2015) later incorporated disparate impact concepts into machine learning fairness research. Its use in machine learning analysis should consequently be interpreted as a screening approach rather than as a direct legal judgement in a credit risk setting.

In this study, the four fifths rule is used only as a transparent screening criterion for favourable outcome rates. A ratio below 0.8 signals a difference requiring further examination rather than proving unlawful discrimination. This interpretation is particularly important because a single group level ratio cannot identify the mechanisms producing an observed disparity.

The broader issue nevertheless remains relevant to lending. Fuster et al. (2022) demonstrate that machine learning approaches can distribute predictive gains unevenly across demographic groups, while Babaei and Giudici (2024) consider fairness measures specifically in credit risk applications. These findings support examining demographic outcomes even where overall predictive performance is strong.

A further challenge is proxy discrimination. Removing a demographic variable does not ensure that correlated information is absent from the model. A low direct attribution to a demographic characteristic therefore does not establish that demographic information is irrelevant to predictions because other correlated features and interactions may still contribute to group level differences. Feature attribution alone is consequently insufficient as a measure of demographic fairness.

Group fairness measures themselves also have theoretical limitations. Chouldechova (2017) demonstrated incompatibilities between fairness properties when outcome prevalence differs between groups, while Kleinberg, Mullainathan, and Raghavan (2017) showed that calibration and particular balance conditions cannot generally be satisfied simultaneously. These results demonstrate that there is no single fairness measure that resolves every fairness concern, reinforcing the need to interpret the four fifths rule according to the specific question it addresses.

For this reason, a four fifths ratio below 0.8 cannot identify the cause of a disparity. Bootstrap confidence intervals are used in the empirical analysis to quantify uncertainty around the estimated ratios, while the ratios themselves are treated as screening evidence rather than causal findings.

Feature attribution and group level fairness therefore provide different forms of evidence. Attribution describes the importance assigned to model features, whereas the four fifths ratio compares favourable outcomes between demographic groups. The two measures may correspond in some cases and diverge in others. Research Question 3 examines this relationship rather than assuming that demographic attribution and group level disparity are equivalent.

2.8 Synthesis and Research Gaps

The literature identifies three related gaps addressed by this study.

First, explanation methods can disagree, but the extent to which concordance varies across model and data conditions remains uncertain. Research Question 1 therefore compares SHAP, LIME and Integrated Gradients using formal rank based measures.

Second, implementation invariance applies to functionally equivalent models but does not establish whether independently trained networks implemented in different frameworks will produce similar explanations. Research Question 2 therefore compares matched TensorFlow and PyTorch networks across multiple random seeds.

Third, fairness metrics and feature attribution describe different aspects of model behaviour and are rarely evaluated together. Research Question 3 compares demographic attribution patterns with favourable outcome disparities measured using the four fifths screening criterion.

The two-dataset design repeats the same broad analytical procedure under contrasting data conditions. This provides comparative evidence beyond a single dataset but does not isolate dataset size from the other differences between the Taiwan and German data.

Differences in data volume and fitted model behaviour are therefore treated as possible explanations for cross dataset patterns rather than predetermined causes. The numerical findings cannot be assumed to generalise universally, but the analytical procedure can be repeated on additional credit datasets to test whether the same patterns persist.

Chapter 3

Methodology

3.1 Research Design

This study investigates the reliability of post hoc explanation methods in credit risk modelling. Predictive performance remains relevant because explanations must be interpreted in the context of the model that produced them, but the principal focus is on the explanations rather than establishing a new credit scoring benchmark.

The three research questions are linked to formal statistical procedures. Research Question 1 compares SHAP (Lundberg and Lee 2017), LIME (Ribeiro, Singh, and Guestrin 2016) and Integrated Gradients (Sundararajan, Taly, and Yan 2017) using Kendall’s coefficient of concordance and pairwise Spearman rank correlations. Research Question 2 compares matched TensorFlow and PyTorch neural networks across multiple random seeds using paired comparisons of SHAP importance and training behaviour. Research Question 3 evaluates demographic outcome disparities using the four fifths screening criterion (Equal Employment Opportunity Commission 1978) and compares these results with demographic feature attribution.

Two credit datasets are used to provide contrasting modelling settings. The Taiwan Default of Credit Card Clients dataset contains 30,000 observations, whereas the German Statlog Credit dataset contains 1,000. They also differ in feature structure and class distribution. The comparison therefore provides evidence beyond a single dataset, although it does not isolate sample size as a causal factor.

Modelling and explanation were conducted in Python using Google Colab, while formal statistical analysis was carried out in R using RStudio. Intermediate results were exported as CSV files so that the statistical analysis could be reproduced independently of the Python notebook state.

Figure 3.1 provides an overview of the complete experimental workflow, showing how the two datasets are linked to the four predictive models, the three explanation

methods and the statistical analyses used to address each research question.

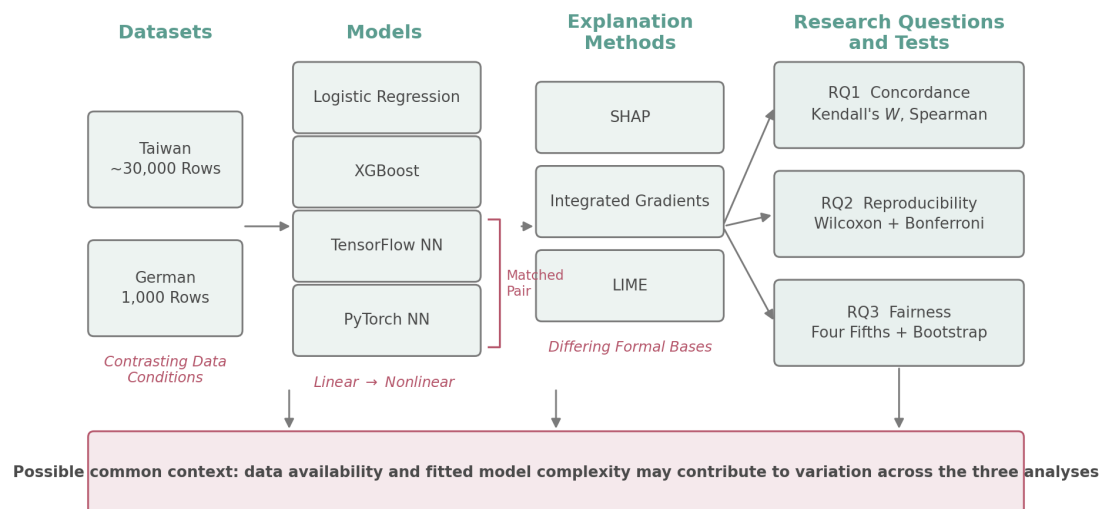


Figure 3.1: Experimental design showing the two datasets, predictive models, explanation methods and statistical analyses used to address the three research questions.

3.2 Datasets

Dataset selection required a binary credit outcome, demographic information for Research Question 3, and public provenance from an identifiable source. Both datasets were obtained from the UCI Machine Learning Repository.

3.2.1 Default of Credit Card Clients Dataset

The UCI Default of Credit Card Clients dataset (Yeh 2009) contains 30,000 observations and 23 original predictor variables. Of these observations, 6,636 belong to the default class, corresponding to a default rate of 22.12%.

The originating study by Yeh and Lien (2009) reports 25,000 observations, including 5,529 defaults (22.12%), whereas the UCI distribution used in this study contains 30,000 observations. The analysis therefore uses and reports the version containing 30,000 observations distributed by the UCI Machine Learning Repository.

Predictors include credit limit, repayment history, bill amounts, payment amounts

and demographic characteristics. Sex, education, marital status and age are included, although the principal group level fairness analysis uses sex and age.

Following preprocessing and one hot encoding, the model input contained 26 encoded features. An 80:20 stratified split produced 24,000 training observations and 6,000 test observations.

3.2.2 Statlog German Credit Dataset

The Statlog German Credit dataset (Hofmann 1994) contains 1,000 observations and 20 original predictor variables. The target was recoded so that bad credit takes the value one, matching the Taiwan adverse outcome convention. Approximately 30% of observations belong to this class.

The dataset is a stratified sample of credits granted between 1973 and 1975. Bad credits were deliberately oversampled, so the 30% adverse outcome rate reflects the sampling design rather than the underlying prevalence of bad credit. The observations also represent borrowers who had already passed the bank’s creditworthiness checks and received credit, introducing selection bias into the population represented by the dataset (Grömping 2019).

The dataset contains a combined personal status and sex field. The legacy UCI documentation was used during the original preprocessing to derive the binary grouping retained in the reporting run. However, subsequent correction of the Statlog coding shows that sex cannot be uniquely recovered from this field because male singles and female non-singles share the same corrected category (Grömping 2019). The resulting German sex variable is therefore treated as a legacy derived proxy rather than a definitive reconstruction of sex. Age was retained numerically.

Following preprocessing and one hot encoding, the model input contained 46 encoded features. The 80:20 stratified split produced 800 training observations and 200 test observations.

Foreign worker status was retained as a predictor but excluded from the principal fairness grouping because the corrected coding identifies only 37 observations as foreign workers, compared with 963 non-foreign workers, producing a highly imbalanced grouping (Grömping 2019).

3.2.3 Applicability to Additional Datasets

The analytical procedure can be reapplied to other tabular binary classification datasets. Dataset specific steps would include identifying the target, separating numerical and categorical predictors, applying any necessary recoding, and identifying demographic attributes where fairness analysis is required.

The subsequent procedure remains unchanged: preprocessing is fitted using training data only, the predictive models are trained, feature attributions are generated, encoded categorical attributions are aggregated to their parent features, and the resulting rankings and predictions are passed to the statistical analyses.

The Taiwan and German datasets demonstrate that the procedure can operate across substantially different sample sizes and feature structures. This does not establish universal generalisability, but provides a reproducible framework that can be tested on additional datasets.

3.3 Data Preprocessing

Preprocessing was kept limited and documented so that transformations did not obscure the relationship between the original variables and their explanations.

For Taiwan, undocumented education categories coded 0, 5 and 6 were combined into the documented “other” category, and the undocumented zero marital status category was treated similarly. Repayment history variables were renamed chronologically, with PAY_1 representing the most recent repayment status.

For German, the binary grouping used in the reporting run was derived from the legacy UCI personal status and sex coding. Subsequent corrected documentation shows that this grouping cannot be interpreted as a definitive reconstruction of sex (Grömping 2019). The target was recoded so that one represents the adverse outcome.

Each dataset was divided using an 80:20 stratified split with random state 42. The split occurred before preprocessing parameters were estimated. Numerical variables were standardised using training set statistics and the same transformation was then applied to the test data.

Categorical variables without a natural ordering were one hot encoded with the

first category dropped. Test columns were aligned to the training representation, and sex was retained as a single binary indicator.

Class imbalance was handled through weighting rather than synthetic oversampling. Logistic regression used balanced class weights, the neural networks used equivalent weighting in their binary cross entropy losses, and XGBoost used the negative to positive class ratio through `scale_pos_weight`. SMOTE (Chawla et al. 2002) was not used because this would introduce synthetic applicant records.

Information Value was calculated as a descriptive diagnostic rather than a feature selection criterion. Continuous variables were divided into quantile based bins where appropriate, while variables with fewer distinct values were treated categorically. Weight of Evidence used smoothing to avoid infinite values in bins containing no good or bad observations.

No predictors were removed on the basis of Information Value. Demographic variables were required for Research Question 3, while univariate Information Value does not capture interactions that nonlinear models may exploit (Siddiqi 2006).

3.4 Predictive Models

Four models were used: logistic regression, XGBoost, and matched feed forward neural networks implemented in TensorFlow/Keras and PyTorch.

Logistic regression provides the linear baseline. XGBoost (Chen and Guestrin 2016) represents the gradient boosted tree family and used 300 estimators, maximum depth four, learning rate 0.05, subsampling of 0.8 and column subsampling of 0.8.

3.4.1 Matched Neural Network Architecture

The TensorFlow and PyTorch networks were matched as closely as practical. Both contained:

- an input layer corresponding to the encoded feature representation;
- a hidden layer of 32 ReLU units;
- dropout with a rate of 0.3;
- a hidden layer of 16 ReLU units; and

- one sigmoid output unit.

The input dimension was 26 for Taiwan and 46 for German. Both implementations used the Adam optimiser (Kingma and Ba 2015) with a learning rate of 0.001, binary cross entropy loss, batch size 128, a maximum of 50 epochs and early stopping with patience eight.

Twenty per cent of the training data was used for validation. Keras used its validation split mechanism, while the corresponding final 20% of the training array was separated manually in PyTorch.

The implementations are therefore matched at the architecture and broad training configuration level but are not assumed to be computationally identical. Framework specific numerical execution and training behaviour remain possible sources of variation.

Class weighting was passed directly to Keras, whereas the PyTorch weighted binary cross entropy implementation was also used when evaluating validation loss. Final validation loss values are consequently treated as implementation diagnostics rather than strictly like for like framework measures.

Random seeds were set to improve repeatability, but TensorFlow operation determinism was not explicitly enabled.

3.4.2 Model Performance Assessment

Predictive performance was assessed primarily using ROC AUC. Five fold stratified cross validation used the same fold assignments for all four models, and fold level values were retained for the Friedman comparison described in Section 3.6.4.

Test set AUC and classification summaries were also calculated as descriptive checks but were not substituted for the matched fold level comparison.

3.5 Explanation Methods

Three feature attribution methods were evaluated: SHAP, Integrated Gradients and LIME.

3.5.1 SHAP

SHAP assigns feature attributions using Shapley values from cooperative game theory (Lundberg and Lee 2017). For feature i ,

$$\phi_i(f, x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (M - |S| - 1)!}{M!} [f_x(S \cup \{i\}) - f_x(S)], \quad (3.1)$$

where $\phi_i(f, x)$ is the attribution assigned to feature i for observation x under model f , N is the complete feature set, $M = |N|$ is the number of features, S is a subset not containing i , and $f_x(S)$ is the model output associated with feature subset S for observation x . The difference $f_x(S \cup \{i\}) - f_x(S)$ is the marginal contribution of feature i . The factorial term weights this contribution across possible feature coalitions.

Logistic regression was explained using SHAP’s LinearExplainer, XGBoost using TreeExplainer, and both neural networks using GradientExplainer. TreeSHAP provides the specialised tree formulation (Lundberg, Erion, et al. 2020).

Training background data were supplied to the linear masker, which internally reduced the effective background to 100 observations in the reporting environment. A fixed background sample of 200 training observations was used for the neural network GradientExplainer.

3.5.2 Integrated Gradients

Integrated Gradients accumulates gradients along a path from a baseline to the input being explained (Sundararajan, Taly, and Yan 2017):

$$IG_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha, \quad (3.2)$$

where $IG_i(x)$ is the attribution for feature i , x is the observed input, x' is the reference baseline, x_i and x'_i are the corresponding observed and baseline feature values, F is the differentiable model function, and α specifies a point along the straight line path from x' to x .

A zero baseline was used for both networks. For standardised numerical variables, zero represents the training set mean; for binary and one hot encoded variables, it

represents the inactive or reference coded state. It therefore does not represent a literal average applicant.

For TensorFlow, Integrated Gradients was implemented using automatic differentiation. With $m = 64$ intervals,

$$IG_i(x) \approx (x_i - x'_i) \frac{1}{m+1} \sum_{k=0}^m \frac{\partial F \left(x' + \frac{k}{m}(x - x') \right)}{\partial x_i}, \quad (3.3)$$

where m is the number of integration intervals and k indexes the $m+1$ path points. With $m = 64$, 65 TensorFlow gradient evaluations were used.

PyTorch Integrated Gradients used Captum with a zero baseline and `n_steps=64`. Integrated Gradients was applied only to the neural networks because it requires a differentiable model.

3.5.3 LIME

LIME explains an observation by fitting an interpretable local surrogate to the original predictive model (Ribeiro, Singh, and Guestrin 2016):

$$\xi(x) = \arg \min_{g \in G} [\mathcal{L}(f, g, \pi_x) + \Omega(g)], \quad (3.4)$$

where $\xi(x)$ is the explanation for observation x , f is the predictive model, G is the family of interpretable surrogate models, g is a candidate surrogate, π_x weights observations according to proximity to x , $\mathcal{L}(f, g, \pi_x)$ measures local surrogate fidelity, and $\Omega(g)$ penalises surrogate complexity.

Tabular LIME was run in classification mode with continuous feature discretisation disabled. Each explanation used 1,000 perturbed samples and returned weights for the adverse class. Model specific wrappers supplied the two class probabilities required by LIME.

3.5.4 Attribution Aggregation and Comparability

One hot encoding means that one original categorical variable may correspond to several encoded columns. Absolute attributions for dummy columns belonging to

the same variable were therefore summed and reassigned to their original parent feature before rankings were calculated.

For original feature j , global importance was defined as

$$I_j = \frac{1}{n} \sum_{i=1}^n |a_{ij}|, \quad (3.5)$$

where I_j is the global importance of feature j , n is the number of explained observations, i indexes observations, and a_{ij} is the attribution assigned to feature j for observation i .

Absolute values were used because the analysis concerns magnitude of influence rather than attribution direction.

Common explanation observations were used across applicable methods wherever possible. The 200 observation German test set was explained in full for the primary RQ1 analysis.

Because explanation methods do not necessarily produce directly interchangeable numerical scales, Research Question 1 compares feature rankings rather than raw attribution magnitudes across methods.

For Research Question 2, fixed explanation and background samples were held constant across all ten seeds so that sampling changes did not confound the framework comparison.

3.6 Statistical Analysis

All formal statistical analyses were conducted in R. Statistical significance was assessed at the 0.05 level unless adjustment for multiple comparisons was required.

3.6.1 RQ1: Explanation Method Agreement

Three way agreement between SHAP, LIME and Integrated Gradients for the neural networks was assessed using Kendall’s coefficient of concordance W (Kendall and Smith 1939). The rankings contained 23 original features for Taiwan and

20 original features for German after encoded categorical attributions had been aggregated to their parent features.

If m explanation methods rank n features and R_i is the rank sum assigned to feature i , the mean rank sum is

$$\bar{R} = \frac{m(n+1)}{2}, \quad (3.6)$$

where $\bar{R}$ is the mean rank sum, m is the number of methods and n is the number of features.

The dispersion of the rank sums is

$$S = \sum_{i=1}^n (R_i - \bar{R})^2, \quad (3.7)$$

where S is the summed squared deviation of the feature rank totals and R_i is the rank sum for feature i .

For ranking j , the tie correction is

$$T_j = \sum_g (t_{jg}^3 - t_{jg}), \quad (3.8)$$

where T_j is the tie correction term, g indexes tied groups and t_{jg} is the number of features in tied group g .

Kendall's coefficient with tie correction is

$$W = \frac{12S}{m^2(n^3 - n) - m \sum_{j=1}^m T_j}, \quad (3.9)$$

where W is Kendall's coefficient of concordance, S is the summed squared deviation defined in Equation (3.7), m is the number of explanation methods, n is the number of ranked features, and T_j is the tie correction term for ranking j defined in Equation (3.8). W ranges from 0 to 1, with larger values indicating stronger agreement.

Formal tie corrected values and significance tests were calculated in R. Pairwise Spearman correlations were also calculated to identify where disagreement was concentrated. Logistic regression and XGBoost support SHAP and LIME but not Integrated Gradients, so their RQ1 analysis uses only the SHAP and LIME Spearman correlation.

3.6.2 RQ2: Framework Comparison and Reproducibility

TensorFlow and PyTorch were compared across ten matched random seeds. The same explanation and background samples were held constant across seeds.

For feature i and seed s ,

$$d_{i,s} = I_{i,s}^{TF} - I_{i,s}^{PT}, \quad (3.10)$$

where $d_{i,s}$ is the paired difference, $I_{i,s}^{TF}$ is TensorFlow mean absolute SHAP importance and $I_{i,s}^{PT}$ is the corresponding PyTorch importance.

Feature level paired differences were assessed using the Wilcoxon signed rank test (Wilcoxon 1945). After zero differences were excluded,

$$W^+ = \sum_{s:d_{i,s}>0} r_s, \quad (3.11)$$

where W^+ is the positive rank statistic, s indexes seeds, $d_{i,s}$ is the paired difference and r_s is the rank assigned to $|d_{i,s}|$.

Bonferroni adjustment was applied across the original features:

$$p_i^{\text{adj}} = \min(qp_i, 1), \quad (3.12)$$

where p_i^{adj} is the adjusted probability value for feature i , p_i is its unadjusted Wilcoxon probability value and q is the number of feature level tests.

With ten paired seeds, the smallest attainable exact two sided Wilcoxon p -value is approximately 0.00195 when all paired differences are nonzero. After Bonferroni adjustment, this corresponds to approximately 0.0449 across the 23 Taiwan features

and 0.0391 across the 20 German features. Statistical significance therefore remained attainable despite the conservative multiple comparison correction.

Adjusted values below 0.05 were treated as statistically significant. Non significance is interpreted as an absence of a detectable difference under this design, not as proof of equivalence.

Per seed test AUC, training epochs and wall clock training time were also compared using paired Wilcoxon procedures. Final validation loss was retained only as an implementation diagnostic because weighting was not identical between frameworks.

Spearman correlation between the mean framework SHAP vectors was used to compare feature ordering, while total mean absolute attribution was reported descriptively to compare overall magnitude. Training time comparisons apply only to the experimental environment.

3.6.3 RQ3: Demographic Outcome Disparity

The favourable outcome was defined as predicted nondefault, corresponding to class 0. Predicted probabilities were converted to class labels using a threshold of 0.5, with probabilities greater than or equal to 0.5 assigned to the adverse class and lower probabilities assigned to nondefault.

For demographic group g ,

$$p_g = \frac{1}{n_g} \sum_{i \in g} \mathbf{1}(\hat{y}_i = 0), \quad (3.13)$$

where p_g is the favourable outcome rate, n_g is the number of observations in group g , i indexes applicants, $\mathbf{1}(\cdot)$ is the indicator function and $\hat{y}_i = 0$ represents predicted nondefault.

For demographic grouping G , the four fifths ratio is

$$R_{4/5} = \frac{\min_{g \in G} p_g}{\max_{g \in G} p_g}, \quad (3.14)$$

where $R_{4/5}$ is the four fifths ratio, G is the set of groups being compared, and the numerator and denominator are respectively the lowest and highest favourable

outcome rates.

A value of 1 indicates equal rates. A point estimate below 0.8 is treated as falling below the screening threshold, not as proof of unlawful discrimination.

Sex and age were examined separately. Taiwan age groups were 21–30, 31–40, 41–50, 51–60 and 61+. German age groups were 19–25, 26–35, 36–45, 46–60 and 61+.

Uncertainty was estimated using 2,000 nonparametric bootstrap resamples (Efron 1979), from which percentile based 95% confidence intervals were calculated.

Demographic attribution was then compared descriptively with the group level results. Outcome disparity does not necessarily imply direct reliance on a demographic feature because correlated predictors, proxies and interactions may also contribute.

3.6.4 Model Performance Comparison

The four models were compared using their five matched cross validation folds. A Friedman test was applied to the ranked AUC values.

Let N denote the number of folds, k the number of models and $\bar{R}_j$ the average rank of model j . The statistic is

$$\chi_F^2 = \frac{12N}{k(k+1)} \left[\sum_{j=1}^k \bar{R}_j^2 - \frac{k(k+1)^2}{4} \right], \quad (3.15)$$

where χ_F^2 is the Friedman statistic, N is the number of matched folds, k is the number of models, j indexes models and $\bar{R}_j$ is the average model rank.

Where the Friedman test was significant, Nemenyi pairwise comparisons were used following Demšar (2006). Because cross validation folds reuse observations across overlapping training sets, the fold level estimates are not fully independent (Dietterich 1998). The resulting Friedman and Nemenyi tests are therefore treated as a supporting comparison of predictive performance rather than as definitive independent tests of model superiority. This comparison provides predictive context for the explanation analysis rather than forming a separate research question.

3.7 Reproducibility and Tools

Both datasets were retrieved directly from the UCI Machine Learning Repository. Separate Python notebooks were used for Taiwan and German to prevent variables or intermediate state being unintentionally shared.

The reporting environment recorded TensorFlow 2.20.0, PyTorch 2.11.0+cpu, scikit-learn 1.6.1 and XGBoost 3.3.0. Formal statistical analysis used R 4.4.1.

Random seeds were set where supported, but TensorFlow operation determinism was not explicitly enabled and complete reruns showed some numerical variation. LIME also uses stochastic perturbation sampling. One complete execution was therefore designated as the primary reporting run, and its CSV files, statistical outputs, tables and figures were kept internally consistent. Separate runs were retained as reproducibility evidence.

Python was used for preprocessing, model fitting and attribution generation. R was used for Kendall concordance, paired Wilcoxon procedures, Bonferroni adjustment, bootstrap confidence intervals, Friedman testing and Nemenyi comparisons. Intermediate CSV files allow the numerical inputs to the statistical analysis to be inspected independently of the Python runtime.

Further reproducibility information, including the execution workflow, software environment, explanation configuration and reporting run procedure, is provided in Appendix A.

The complete code and supporting analysis files are available at:

<https://github.com/GDavis-01/Explainable-AI-for-Credit-Risk-Assessment>

The repository contains the Python notebooks, R analysis code, intermediate CSV files and supporting documentation required to reproduce or extend the analytical workflow.

All analytical figures were generated programmatically from the analysis outputs. No analytical figure was manually altered to change the numerical results.

Chapter 4

Results

4.1 Overview

This chapter presents the statistical and descriptive results organised around the three research questions, together with a supporting comparison of predictive performance. Unless otherwise stated, numerical results, tables and analytical figures are taken from the designated complete reporting run described in Section 3.7. Broader interpretation is reserved for Chapter 5.

4.2 Model Performance

4.2.1 Cross Validated Performance

As a supporting comparison of predictive performance, Table 4.1 reports the mean and standard deviation of the five fold cross validated AUC for each model and dataset.

Table 4.1: Five fold cross validated AUC, reported as mean with standard deviation in parentheses.

Model	Taiwan	German
Logistic regression	0.7276 (0.0127)	0.7685 (0.0622)
XGBoost	0.7829 (0.0076)	0.7852 (0.0466)
TensorFlow NN	0.7763 (0.0102)	0.7481 (0.0368)
PyTorch NN	0.7776 (0.0078)	0.7136 (0.0941)

On Taiwan, XGBoost achieved the highest mean AUC at 0.7829, followed closely by PyTorch and TensorFlow at 0.7776 and 0.7763. Logistic regression produced the lowest mean at 0.7276. Variation between folds was small.

On German, XGBoost again produced the highest mean at 0.7852. Logistic regression produced 0.7685, TensorFlow produced 0.7481, and PyTorch produced 0.7136. Fold to fold variation was greater than on Taiwan, particularly for PyTorch.

Figure 4.1 compares the fold level AUC distributions across the four models and highlights the greater variability observed on the German dataset, particularly for the PyTorch network.

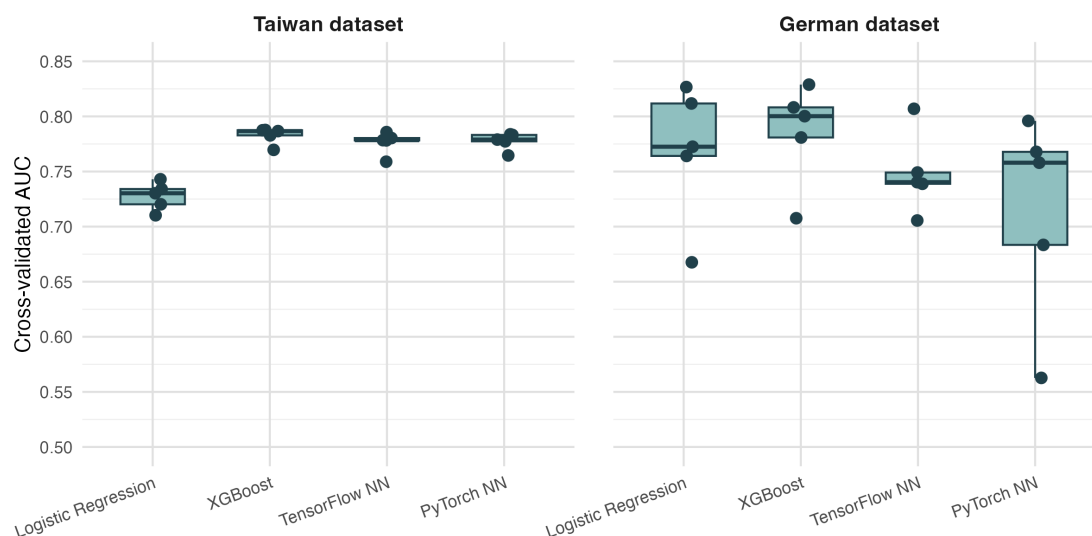


Figure 4.1: Fold level cross validated AUC for the four models on each dataset. Boxes show the interquartile range, with individual fold values shown by the plotted observations. The Taiwan estimates are tightly grouped, whereas the German results show greater fold to fold variability, particularly for the PyTorch network.

4.2.2 Statistical Comparison of Model Performance

For the supporting comparison of predictive performance, the matched fold level AUC scores were compared using the Friedman and Nemenyi procedures described in Section 3.6.4.

For Taiwan, the Friedman test identified a statistically significant difference between the four models, with $\chi_F^2 = 13.56$, $df = 3$, and $p = 0.0036$. The only significant Nemenyi comparison was XGBoost against logistic regression, with $p = 0.0014$. TensorFlow and PyTorch did not differ significantly, with $p = 0.9948$.

For German, the Friedman test was not statistically significant, with $\chi_F^2 = 7.08$, $df = 3$, and $p = 0.0694$. No post hoc comparison was therefore performed.

4.3 Explanation Method Agreement (RQ1)

4.3.1 Overall Concordance

For RQ1, overall agreement between SHAP, LIME and Integrated Gradients for the neural networks was assessed using tie corrected Kendall’s W .

Table 4.2: Tie corrected three way Kendall’s W for SHAP, LIME and Integrated Gradients on the neural networks.

Dataset	Model	Kendall’s W	p
Taiwan	TensorFlow	0.605	0.0109
Taiwan	PyTorch	0.730	0.0010
German	TensorFlow	0.944	3.52×10^{-5}
German	PyTorch	0.935	4.27×10^{-5}

All four coefficients were statistically significant. Statistical significance indicates evidence of concordance in the feature rankings, but does not by itself imply that the explanation methods are interchangeable or closely aligned. The magnitude of W and the pairwise correlations must therefore also be considered. Overall concordance was higher on German than on Taiwan for both neural network implementations.

4.3.2 Pairwise Agreement

For RQ1, pairwise Spearman correlations identify which explanation method combinations account for the differences within the overall concordance.

On Taiwan, SHAP and Integrated Gradients agree strongly for TensorFlow and PyTorch, with correlations of 0.890 and 0.952. Correlations involving LIME are substantially lower, reaching 0.265 and 0.069 for TensorFlow and 0.418 and 0.414 for PyTorch.

Table 4.3: Pairwise Spearman rank correlations between explanation methods.

Dataset	Model	SHAP and IG	SHAP and LIME	IG and LIME
Taiwan	Logistic regression	N/A	0.938	N/A
Taiwan	XGBoost	N/A	0.587	N/A
Taiwan	TensorFlow NN	0.890	0.265	0.069
Taiwan	PyTorch NN	0.952	0.418	0.414
German	Logistic regression	N/A	0.967	N/A
German	XGBoost	N/A	0.848	N/A
German	TensorFlow NN	0.905	0.964	0.880
German	PyTorch NN	0.893	0.965	0.847

On German, agreement involving LIME is stronger. SHAP and LIME correlations are 0.964 for TensorFlow and 0.965 for PyTorch, while Integrated Gradients and LIME correlations are 0.880 and 0.847. SHAP and Integrated Gradients agreement remains high at 0.905 and 0.893.

Figure 4.2 provides a visual comparison of the pairwise explanation method correlations, highlighting the weaker agreement involving LIME on the Taiwan neural networks and the stronger agreement observed on German.

4.3.3 Feature Rankings and Overlap

For RQ1, feature level agreement was examined through feature rankings and overlap, with PAY_1, representing the most recent repayment status, identified as the highest ranked Taiwan SHAP feature for all four models. Its mean absolute SHAP value is 0.4023 for logistic regression, 0.4646 for XGBoost, 0.0958 for TensorFlow and 0.0945 for PyTorch.

The logistic regression ranking differs from the nonlinear models in the prominence of demographic variables, with education ranked third and marital status fifth.

Figure 4.3 compares the twelve highest ranked XGBoost SHAP features across the two datasets, allowing the dominant behavioural, financial and demographic predictors in each modelling setting to be examined directly.

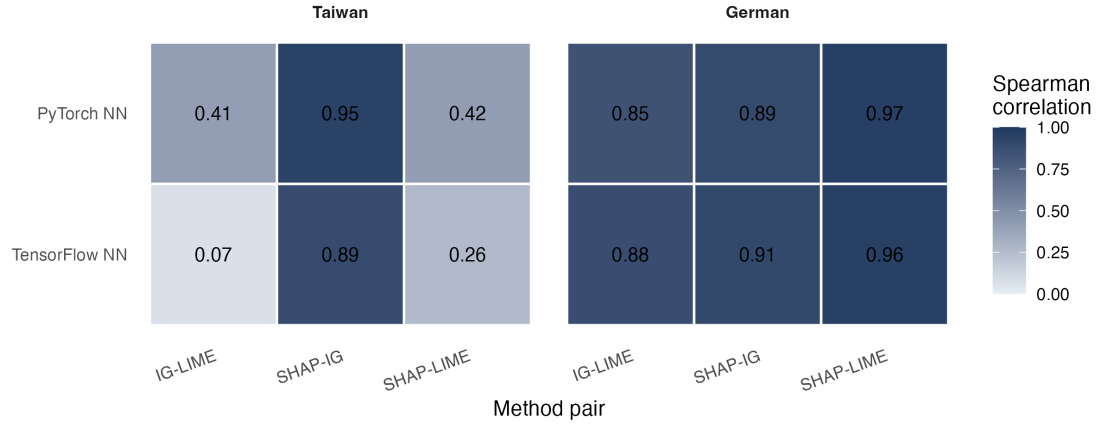


Figure 4.2: Pairwise Spearman rank correlations between SHAP, Integrated Gradients and LIME for the TensorFlow and PyTorch neural networks. Correlations involving LIME are substantially lower on the Taiwan networks than on the German networks.

Table 4.4: Five highest ranked Taiwan features by mean absolute SHAP importance for each model.

Rank	Logistic regression	XGBoost	TensorFlow NN	PyTorch NN
1	PAY_1	PAY_1	PAY_1	PAY_1
2	BILL_AMT1	LIMIT_BAL	LIMIT_BAL	LIMIT_BAL
3	EDUCATION	BILL_AMT1	BILL_AMT2	PAY_3
4	LIMIT_BAL	PAY_2	PAY_3	BILL_AMT2
5	MARRIAGE	PAY_AMT2	PAY_4	PAY_4

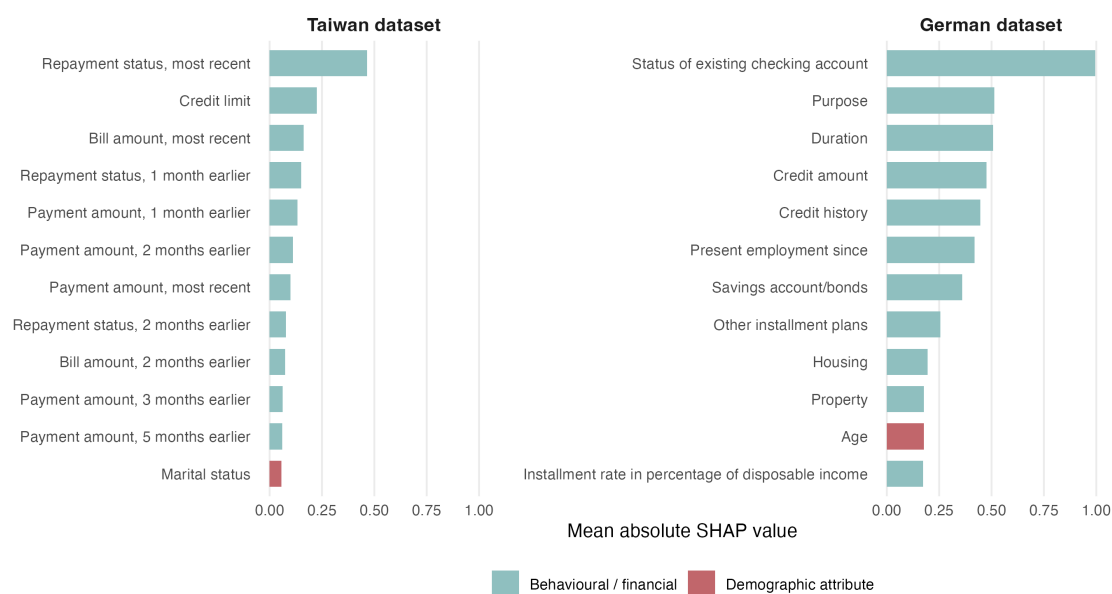


Figure 4.3: Twelve highest ranked features by mean absolute SHAP value for the XGBoost model on the Taiwan and German datasets. Taiwan feature codes are replaced with descriptive labels for readability, and demographic attributes appearing within the top twelve are highlighted separately. Features are ordered by mean absolute SHAP importance rather than chronological sequence.

Behavioural and financial variables account for most of the leading XGBoost features. The Taiwan ranking is concentrated around repayment history, payment behaviour and credit exposure, while the German ranking includes account status, loan characteristics and credit history.

Table 4.5 compares the five highest ranked SHAP and LIME features.

Table 4.5: Overlap between the five highest ranked features identified by SHAP and LIME.

Dataset	Logistic regression	XGBoost	TensorFlow NN	PyTorch NN
Taiwan	3 of 5	4 of 5	2 of 5	2 of 5
German	5 of 5	4 of 5	4 of 5	4 of 5

The lowest overlap occurs for the Taiwan neural networks, where only two of the five highest ranked features are shared between SHAP and LIME.

4.4 Framework Comparison and Reproducibility (RQ2)

4.4.1 Run to Run Reproducibility

For RQ2, repeated complete executions were examined to assess reproducibility beyond the formal ten seed TensorFlow and PyTorch comparison. The designated reporting run was fixed before the supporting reruns were compared and was not selected on the basis of producing a particular concordance result.

For Taiwan, the designated TensorFlow result was $W = 0.605$, while four separate complete runs produced approximately 0.649, 0.707, 0.652 and 0.618. The designated value was therefore the lowest of the five observed complete runs, which ranged from approximately 0.605 to 0.707. The PyTorch result remained close to 0.730. For German, the designated TensorFlow result was $W = 0.944$, compared with approximately 0.906, 0.874, 0.885 and 0.899 in the separate runs, while PyTorch remained close to 0.935.

TensorFlow operation determinism was not explicitly enabled, and correlations involving LIME could also move because of stochastic perturbation sampling.

The separate runs are therefore retained as reproducibility evidence rather than combined with the designated reporting run values.

4.4.2 Feature Level Framework Comparison

For RQ2, feature level TensorFlow and PyTorch SHAP differences were assessed across ten matched seeds, with none of the 23 Taiwan features reaching statistical significance after Bonferroni correction.

For German, two of the 20 original features remained significant.

Table 4.6: German features with statistically significant TensorFlow and PyTorch SHAP differences after Bonferroni correction.

Feature	TensorFlow	PyTorch	Adjusted p
Present residence since	0.0135	0.0084	0.0391
Savings account/bonds	0.0378	0.0558	0.0391

The remaining 18 German features and all 23 Taiwan features did not reach the corrected 0.05 significance level. These results indicate an absence of statistically detectable differences under the ten seed comparison rather than proof of framework equivalence.

4.4.3 Attribution Scale and Feature Ranking

For RQ2, attribution scale and feature ordering were compared between TensorFlow and PyTorch, with Taiwan total mean absolute SHAP attribution of 0.4134 for TensorFlow and 0.4200 for PyTorch, giving a TensorFlow to PyTorch ratio of 0.984. The corresponding feature ranking Spearman correlation is 0.957.

Figure 4.4 compares the leading TensorFlow and PyTorch SHAP attributions on Taiwan, showing the close correspondence in feature ordering and attribution magnitude between the two frameworks.

The repayment status variables are labelled chronologically for clarity, but their positions in Figure 4.4 reflect mean SHAP importance rather than month.

On German, total mean absolute SHAP attribution is 0.6437 for TensorFlow

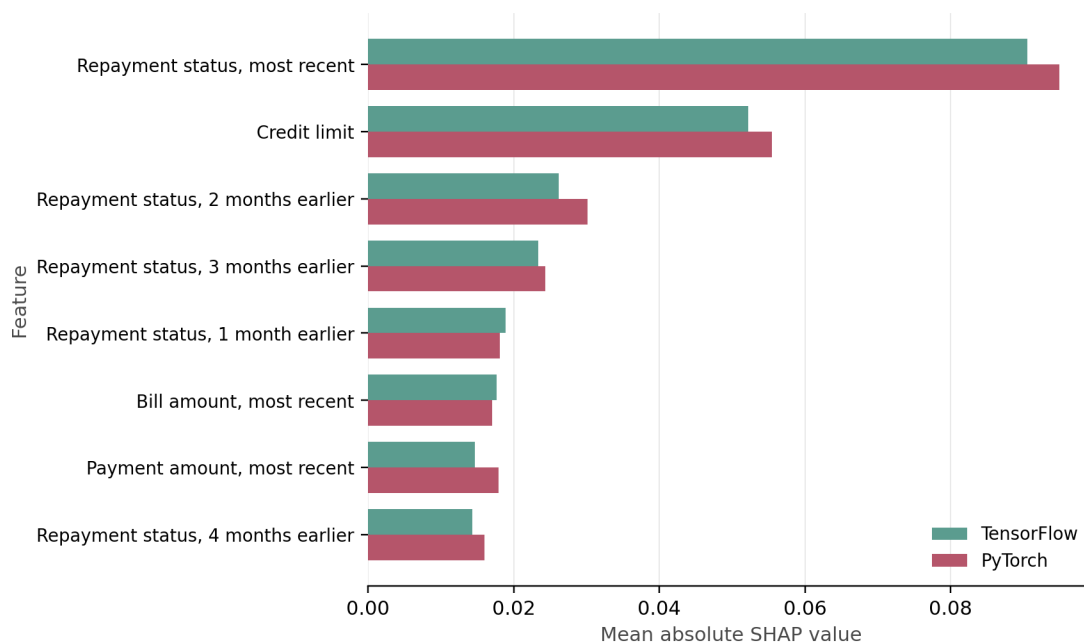


Figure 4.4: Mean absolute SHAP values for the eight most influential features in the TensorFlow and PyTorch models on the Taiwan dataset, averaged across ten seeds. Features are selected and ordered according to their mean importance across the two frameworks rather than chronologically.

and 0.7717 for PyTorch, giving a ratio of 0.834. The feature ranking Spearman correlation is 0.962.

Figure 4.5 compares the leading TensorFlow and PyTorch SHAP attributions on German, showing similar feature ordering despite a larger difference in overall attribution magnitude than was observed on Taiwan.

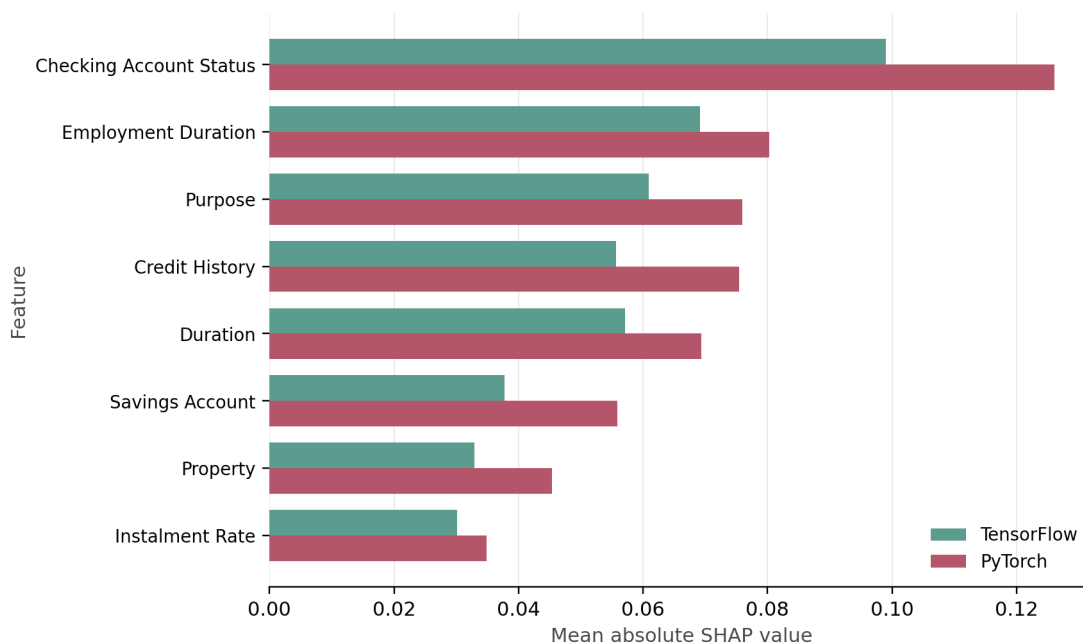


Figure 4.5: Mean absolute SHAP values for the eight most influential German features in the TensorFlow and PyTorch models, averaged across ten seeds. Features are selected and ordered according to their mean importance across the two frameworks.

Feature ordering is therefore strongly aligned on both datasets, while the difference in overall attribution magnitude is larger on German.

4.4.4 Training Behaviour

For the training behaviour component of RQ2, Table 4.7 summarises the paired TensorFlow and PyTorch comparison across ten seeds.

Epoch counts are almost identical and do not differ significantly. Test AUC differs significantly for both datasets, although the absolute difference is small on Taiwan at approximately 0.0021 and larger on German at approximately 0.0471.

Table 4.7: Mean TensorFlow and PyTorch training outcomes across ten seeds, with paired Wilcoxon p values.

Dataset	Measure	TensorFlow	PyTorch	p
Taiwan	Test AUC	0.7711	0.7732	0.0098
Taiwan	Epochs	41.5	41.9	0.8885
Taiwan	Training time (s)	20.39	17.66	0.1055
German	Test AUC	0.7381	0.7852	0.0273
German	Epochs	37.9	37.7	0.9188
German	Training time (s)	4.91	0.54	0.0020

Mean training time was lower for PyTorch on both datasets. PyTorch was approximately 1.15 times faster on Taiwan and 9.09 times faster on German based on the mean recorded times. The Taiwan training time difference was not statistically significant, with $p = 0.1055$, whereas the German difference was statistically significant, with $p = 0.0020$. These timing results are specific to the reported runtime and are not hardware independent benchmarks.

Final validation loss was retained as an implementation diagnostic rather than a primary comparison because the two frameworks did not calculate the weighted validation loss identically.

4.5 Demographic Outcome Disparity (RQ3)

4.5.1 Taiwan Dataset

For RQ3, Table 4.8 reports the Taiwan demographic outcome disparities using four fifths ratios and 95% bootstrap confidence intervals.

The clearest Taiwan result is the logistic regression age ratio of 0.680, whose full confidence interval remains below 0.8. Its sex ratio is 0.799, but the confidence interval crosses the threshold.

The other three models have sex and age point estimates above 0.8, although their age confidence intervals extend below the threshold.

Table 4.8: Taiwan four fifths ratios with 95% bootstrap confidence intervals. The favourable outcome is predicted nondefault.

Model	Sex ratio	Age ratio
Logistic regression	0.799 [0.764, 0.835]	0.680 [0.490, 0.762]
XGBoost	0.926 [0.896, 0.959]	0.860 [0.673, 0.934]
TensorFlow NN	0.948 [0.917, 0.978]	0.857 [0.685, 0.898]
PyTorch NN	0.918 [0.886, 0.949]	0.888 [0.702, 0.949]

Note: Age-band sample sizes in the test set were 21–30 ($n = 2,181$), 31–40 ($n = 2,105$), 41–50 ($n = 1,243$), 51–60 ($n = 413$) and 61+ ($n = 58$).

4.5.2 German Dataset

For RQ3, Table 4.9 reports the corresponding German demographic outcome disparities using four fifths ratios and 95% bootstrap confidence intervals.

Table 4.9: German four fifths ratios with 95% bootstrap confidence intervals. The favourable outcome is predicted nondefault.

Model	Sex ratio	Age ratio
Logistic regression	0.861 [0.637, 0.991]	0.627 [0.357, 0.757]
XGBoost	0.907 [0.699, 0.995]	0.663 [0.440, 0.782]
TensorFlow NN	0.705 [0.491, 0.938]	0.485 [0.278, 0.651]
PyTorch NN	0.640 [0.452, 0.868]	0.473 [0.276, 0.628]

Note: Age-band sample sizes in the test set were 19–25 ($n = 44$), 26–35 ($n = 66$), 36–45 ($n = 45$), 46–60 ($n = 32$) and 61+ ($n = 13$).

All four German age point estimates fall below 0.8, and every corresponding confidence interval also remains below the threshold.

For sex, logistic regression and XGBoost have point estimates above 0.8. TensorFlow and PyTorch produce point estimates below 0.8, at 0.705 and 0.640 respectively, although both confidence intervals cross the threshold.

4.5.3 Worked Example: Taiwan Age Outcomes

To illustrate the RQ3 group level calculation, Taiwan logistic regression provides a worked example in which the favourable outcome rate falls from 0.659 for applicants aged 21–30 to 0.448 for those aged 61 and above, giving $0.448/0.659 \approx 0.680$.

Figure 4.6 compares favourable outcome rates across the Taiwan age bands and shows that the decline is most pronounced for logistic regression, rather than being confined to a single age group.

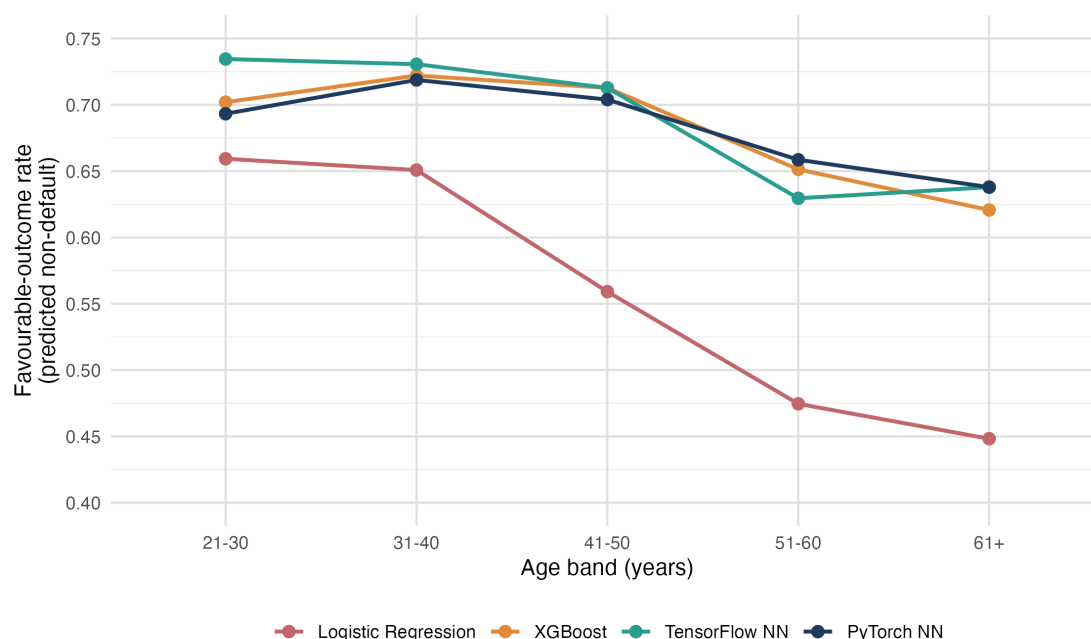


Figure 4.6: Taiwan favourable outcome rates by age band for each predictive model. The logistic regression favourable outcome rate declines from 0.659 for applicants aged 21–30 to 0.448 for those aged 61 and above, producing a four fifths ratio of 0.680.

The intermediate logistic regression rates are 0.651 for ages 31–40, 0.559 for ages 41–50 and 0.475 for ages 51–60. The disparity therefore develops progressively across the age bands rather than being driven by one isolated group.

4.5.4 Relationship Between Attribution and Outcome Disparity

To address the second part of RQ3, Table 4.10 reports mean absolute SHAP importance for the Taiwan demographic features for comparison with the group level outcome disparities.

Table 4.10: Mean absolute SHAP importance assigned to Taiwan demographic features.

Feature	Logistic regression	XGBoost	TensorFlow NN	PyTorch NN
Age	0.0519	0.0409	0.0076	0.0079
Education	0.1146	0.0480	0.0136	0.0181
Marital status	0.0907	0.0571	0.0137	0.0142
Sex	0.0599	0.0504	0.0128	0.0107

Logistic regression assigns substantially greater SHAP importance to education and marital status than the two neural networks. The attribution and group level outcome measures therefore provide related but not identical information.

Figure 4.7 compares the sex and age four fifths point estimates across both datasets and shows which model and demographic combinations fall below the 0.8 screening threshold.

Figure 4.7 shows that the Taiwan logistic regression sex and age ratios, all four German age ratios, and the German TensorFlow and PyTorch sex ratios fall below the screening threshold. The Taiwan logistic regression sex confidence interval and both German neural network sex confidence intervals cross the threshold.

4.6 Summary

RQ1 shows statistically significant overall concordance for all four neural network analyses, but pairwise agreement differs by method and dataset. SHAP and Integrated Gradients remain strongly correlated, while the weakest agreement involves LIME on the Taiwan neural networks.

RQ2 shows strong TensorFlow and PyTorch feature ordering agreement, with

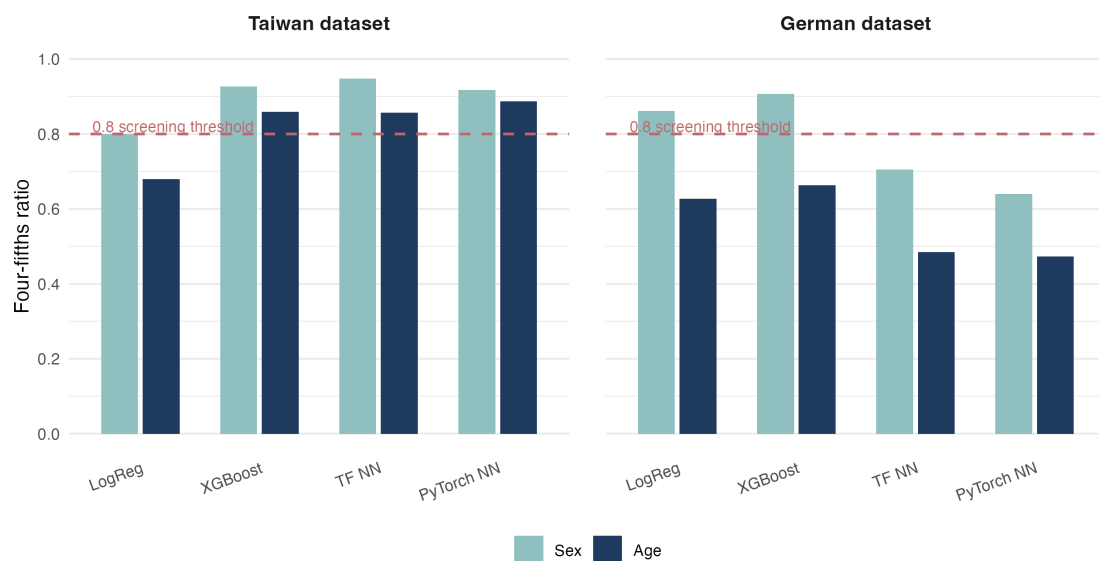


Figure 4.7: Four fifths point estimates for sex and age by model and dataset. The dashed horizontal line marks the 0.8 screening threshold. Bootstrap confidence intervals are reported separately in Tables 4.8 and 4.9.

Spearman correlations of 0.957 on Taiwan and 0.962 on German. No Taiwan features and two German features differ significantly after Bonferroni correction. RQ3 identifies age as the most consistent demographic outcome disparity, particularly for Taiwan logistic regression and across all four German models.

These empirical patterns are interpreted in Chapter 5.

Chapter 5

Discussion

5.1 Overview

The results show that post hoc explanation reliability has several distinct dimensions. The three research questions examine agreement between explanation methods, reproducibility across independently trained software implementations, and the relationship between feature attribution and demographic outcome disparities. Taken together, the findings show that reliability cannot be inferred from a single measure or explanation output.

Differences between the Taiwan and German results are consistent with the possibility that data availability and fitted model complexity influence explanation behaviour. However, the two datasets differ in several respects, so neither dataset size nor model nonlinearity can be isolated as the cause of these patterns. The cross dataset comparison therefore provides evidence of variation between modelling settings rather than a causal explanation for that variation.

5.2 RQ1: Explanation Method Agreement Is Conditional

Research Question 1 asked whether SHAP, LIME and Integrated Gradients produce similar global feature rankings. The overall Kendall's W results were statistically significant for all four neural network analyses, but the pairwise correlations show that this overall agreement conceals substantial differences between methods. Explanation agreement should therefore be considered at both the overall and pairwise levels.

SHAP and Integrated Gradients show consistently strong agreement, with Spearman correlations ranging from 0.890 to 0.952. In contrast, the weakest agreement

occurs in comparisons involving LIME on the Taiwan neural networks. SHAP and LIME correlations are 0.265 for TensorFlow and 0.418 for PyTorch, while Integrated Gradients and LIME correlations are 0.069 and 0.414. On German, the corresponding LIME comparisons are substantially stronger, while logistic regression also shows high SHAP and LIME agreement on both datasets.

These results demonstrate why overall concordance should not be interpreted in isolation. Statistical significance of Kendall’s W indicates evidence of concordance in the rankings, but the magnitude of W is required to judge the strength of that agreement. A statistically significant W does not mean that every pair of explanation methods produces closely matching rankings or that the methods are interchangeable. A model audit based only on the overall statistic could therefore conceal disagreement between individual explanation methods.

One possible explanation for the LIME pattern lies in its construction. LIME approximates model behaviour locally using a linear surrogate (Ribeiro, Singh, and Guestrin 2016). Where the local decision surface is poorly represented by a linear approximation, its ranking may differ from those generated by other methods. The close SHAP and LIME agreement for logistic regression and weaker agreement for the Taiwan neural networks are consistent with this explanation because a linear surrogate may more closely approximate an already linear model.

However, model nonlinearity was not measured directly, so the results do not establish it as the cause of LIME disagreement. Dataset size alone is also insufficient as an explanation because SHAP and LIME agree strongly for logistic regression on both datasets despite their substantial difference in sample size. The observed disagreement is therefore more plausibly interpreted as dependent on a combination of the explanation method, fitted model and data setting rather than any single factor.

The strong SHAP and Integrated Gradients agreement also requires qualification. Neural network SHAP was produced using a gradient based explainer, meaning that both approaches use gradient information from the fitted differentiable model. Their agreement may therefore reflect some shared methodological information rather than completely independent convergence (Lundberg and Lee 2017; Sundararajan, Taly, and Yan 2017). Strong agreement between these methods is useful evidence of consistency, but it should not be interpreted as confirmation from two entirely independent explanatory approaches.

SHAP implementation also differed between model families, with linear, tree based and gradient based explainers used according to model type. Comparisons of SHAP behaviour across model families may therefore reflect both differences in the fitted models and differences in the explainer formulations, which cannot be separated completely in the present design.

RQ1 consequently supports a conditional conclusion: explanation methods can show strong overall concordance while still differing substantially under particular combinations of method, model and dataset. Explanation agreement should therefore be tested within the modelling setting in which the explanation is intended to be used.

5.3 RQ2: Framework Reproducibility Is High but Not Complete

Research Question 2 examined whether matched TensorFlow and PyTorch implementations produce similar feature attribution patterns and training behaviour. The findings indicate strong reproducibility in feature ordering, but weaker evidence for numerical equivalence between independently trained implementations.

The strongest evidence of reproducibility concerns feature ordering. TensorFlow and PyTorch SHAP rankings correlate at 0.957 on Taiwan and 0.962 on German. No Taiwan features differ significantly after Bonferroni correction, while two of the 20 German features do. The high correlations therefore indicate that the frameworks generally identify similar features as important, even though the individual attribution values are not identical.

These results are not proof of framework equivalence. Only ten paired seeds were available and Bonferroni correction is conservative, meaning that smaller differences may remain undetected. Formal equivalence testing would be required to support a stronger claim that framework differences are practically negligible. The results therefore support high reproducibility rather than interchangeability.

The attribution totals further distinguish ranking agreement from numerical agreement. Taiwan total mean absolute SHAP attribution is 0.4134 for TensorFlow and 0.4200 for PyTorch, giving a ratio of 0.984. On German, the corresponding values are 0.6437 and 0.7717, giving a ratio of 0.834. The frameworks can therefore

identify the same variables as relatively important while assigning different overall attribution magnitudes. This demonstrates that agreement in ordering does not imply agreement in the strength of the explanations.

The comparison relates to, but is not a direct test of, implementation invariance (Sundararajan, Taly, and Yan 2017). That principle concerns models computing the same mathematical function. The TensorFlow and PyTorch networks were independently trained implementations of a matched architecture and were not guaranteed to learn identical functions. The experiment therefore evaluates practical cross framework reproducibility rather than theoretical implementation invariance.

Training behaviour provides further evidence of this distinction. The ten seed mean test AUCs are close on Taiwan at 0.7711 and 0.7732, although the paired difference is statistically significant. The German difference is larger, with means of 0.7381 and 0.7852. However, the German cross validated AUC estimates also show substantially greater fold to fold variability than Taiwan, as shown in Figure 4.1, so this larger framework difference should be interpreted in the context of greater performance uncertainty on the smaller dataset. Epoch counts do not differ significantly on either dataset. These results show that matched architectures and training settings can still produce measurable differences between implementations.

Mean training time is lower for PyTorch within the reported Google Colab environment, by approximately 1.15 times on Taiwan and 9.09 times on German. The Taiwan training time difference is not statistically significant, with $p = 0.1055$, whereas the German difference is statistically significant, with $p = 0.0020$. The much larger German timing ratio likely reflects fixed framework and per epoch overhead having a greater influence on the smaller 800 observation training set, rather than demonstrating a general PyTorch speed advantage. These values are environment specific and should not be interpreted as general framework benchmarks. Final validation loss is similarly retained only as an implementation diagnostic because the weighting procedures were not directly comparable. These measures therefore provide information about the reported experimental environment rather than evidence that one framework is generally superior.

Repeated complete executions provide a further qualification. TensorFlow derived concordance values showed some movement despite nominally identical seeds, whereas PyTorch model derived SHAP and Integrated Gradients outputs were more stable under the seeding procedure. LIME itself also introduces stochastic

variation. TensorFlow operation determinism was not explicitly enabled, so the contribution of framework execution and explanation stochasticity cannot be separated completely.

RQ2 therefore indicates high cross framework reproducibility in feature ordering, but not complete numerical invariance. Reproducibility should consequently be evaluated using more than predictive performance alone when explanations form part of the model auditing process.

5.4 RQ3: Demographic Outcome Disparity and Attribution Provide Complementary Evidence

Research Question 3 examined demographic outcome disparities using the four fifths screening criterion and compared these results with demographic feature attribution. The findings show that attribution and outcome level disparity can correspond, but they measure different aspects of model behaviour and should not be treated as equivalent.

Age produces the clearest pattern. For Taiwan logistic regression, the age ratio is 0.680 with a 95% bootstrap confidence interval of 0.490 to 0.762, placing both the estimate and full interval below 0.8. The remaining Taiwan models have age point estimates above the threshold, although their confidence intervals extend below it.

The German results are more consistent. All four age ratios fall below 0.8, ranging from 0.473 for PyTorch to 0.663 for XGBoost, and the upper confidence limit remains below the threshold in every case. Sex produces less consistent evidence. Taiwan logistic regression has a point estimate of 0.799 but an interval crossing 0.8. On German, TensorFlow and PyTorch have point estimates of 0.705 and 0.640 respectively, with both confidence intervals crossing the threshold.

These results show why uncertainty should accompany the four fifths point estimate. A value slightly below 0.8 with a confidence interval crossing the threshold should not be interpreted in the same way as a result whose full interval remains below it (Efron 1979). The bootstrap intervals therefore add important context to a screening rule that might otherwise encourage an overly binary interpretation.

Feature attribution provides complementary rather than equivalent evidence. Tai-

wan logistic regression assigns greater SHAP importance to several demographic variables than the neural networks and also produces the clearest Taiwan age disparity. Greater demographic attribution and outcome disparity can therefore occur together, suggesting that direct attribution may provide useful evidence about variables that warrant further investigation.

However, XGBoost demonstrates that the relationship is not direct. Its Taiwan age attribution remains nonzero while its age point estimate is 0.860. Group level disparities may also arise through correlated variables, proxies and interactions without a demographic variable receiving a dominant direct attribution. A low demographic attribution should therefore not be interpreted as evidence that demographic disparity is absent.

Attribution should consequently not replace formal fairness assessment. Attribution measures the importance assigned to variables within a model explanation, whereas the four fifths ratio measures differences in favourable outcomes between groups. The two approaches answer different questions, and their value lies in considering them together rather than treating one as a substitute for the other.

The four fifths criterion itself also requires caution. It originates in United States employment selection guidance (Equal Employment Opportunity Commission 1978) and is used here as a screening measure rather than a lending specific legal test. A ratio below 0.8 does not prove unlawful discrimination or identify its cause. Its role in this study is therefore to identify outcome differences that warrant further examination.

This reflects the wider fairness literature, which shows that different fairness criteria represent different objectives and cannot always be satisfied simultaneously (Chouldechova 2017; Kleinberg, Mullainathan, and Raghavan 2017). The choice of fairness measure should therefore be matched to the particular question being investigated rather than interpreted as a universal definition of fairness.

The regulatory relevance is broader than the threshold itself. UK data protection legislation retains safeguards around significant automated decisions (United Kingdom 2025), while the FCA Consumer Duty places emphasis on consumer understanding (Financial Conduct Authority 2022). In the European Union, specified creditworthiness and credit scoring AI systems are categorised as high risk under the Artificial Intelligence Act, subject to its scope and exceptions (European Union

2024). These frameworks differ in legal effect, but each reinforces the importance of meaningful scrutiny of consequential automated systems.

RQ3 therefore supports using feature attribution and outcome level fairness measures as complementary forms of evidence. Neither alone provides a complete account of demographic model behaviour.

5.5 A Proposed Unifying Account of the Findings

The three research questions suggest a possible common context involving data availability and fitted model complexity. This is proposed as an interpretation of the cross dataset pattern rather than as a causal result.

Taiwan contains substantially more observations and is the setting in which LIME diverges most strongly from the neural network gradient based explanations. It is also the only dataset where the Friedman test detects an overall model performance difference, with $\chi_F^2 = 13.56$ and $p = 0.0036$. On German, the corresponding test is not significant, with $\chi_F^2 = 7.08$ and $p = 0.0694$, while fold level variability is greater.

These findings are consistent with the possibility that a larger sample allows flexible models to learn relationships that are more difficult for a local linear surrogate to approximate. Conversely, a smaller sample may permit greater variation between independently trained implementations. This provides one possible account of why the strongest LIME disagreement occurs on Taiwan while larger framework level attribution differences are observed on German.

However, neither mechanism was measured directly. Dataset size was not experimentally manipulated, model nonlinearity was not quantified, and Taiwan and German also differ in feature composition, class distribution and population. It would therefore be inappropriate to conclude that data volume caused the observed differences. The proposed account should instead be treated as a hypothesis that can be tested more directly in future work.

The fairness findings provide an additional reason for caution. German age disparities occur across all four models despite demographic attributes not dominating the feature rankings. Fairness patterns cannot therefore be reduced to the same explanation used for attribution concordance or framework reproducibility.

The narrower conclusion is that explanation agreement, implementation reproducibility, predictive stability and demographic outcome disparities vary across data and modelling conditions. Data availability and model complexity are plausible contributors, but additional controlled experiments are required before either can be treated as a general explanation.

5.6 Comparison with Related Work

The predictive results are broadly consistent with existing credit scoring research. XGBoost achieves the highest mean cross validated AUC on both datasets, consistent with evidence that ensemble methods perform strongly on structured credit data (Lessmann et al. 2015). The neural networks do not show a consistent advantage over XGBoost, which also agrees with previous comparisons of deep learning and gradient boosting (Gunnarsson et al. 2021). The present results therefore provide further evidence that additional model complexity does not necessarily translate into stronger credit scoring performance.

The explanation findings help reconcile differing conclusions in the literature. Gramegna and Giudici (2021) reported useful agreement between SHAP and LIME in credit risk modelling, while Krishna et al. (2022) documented substantial disagreement between explanation methods across models and datasets. These findings could appear contradictory if explanation agreement is assumed to be a fixed property of the methods.

Both patterns occur in the present study. SHAP and LIME agree strongly for logistic regression and for the German neural networks but diverge substantially for the Taiwan neural networks. The results therefore support a conditional interpretation of the previous literature: explanation agreement can be high in some settings and weak in others. Agreement should consequently be measured within the model and data setting being audited rather than assumed from evidence obtained elsewhere.

The fairness findings similarly support using multiple forms of evidence. Machine learning lending models can produce different effects across demographic groups (Fuster et al. 2022), while explanations may contribute to model auditability (Bücker et al. 2022). The present results demonstrate that attribution and outcome level fairness measures can corroborate one another but are not interchangeable. This

reinforces the value of combining explanation analysis with a separate assessment of demographic outcomes.

5.7 Implications for Practice

The findings suggest four practical considerations.

1. **Explanation agreement should be checked rather than assumed.**

The Taiwan neural network results show that explanation method choice can materially alter global feature rankings. Comparing methods with different theoretical foundations can therefore reveal explanation uncertainty that would be hidden by using only one method.

2. **Flexible models may warrant greater explanation scrutiny.**

SHAP and LIME agree strongly for logistic regression on both datasets, while their greatest disagreement occurs for the Taiwan neural networks. Although nonlinearity is not established as the cause, explanation reliability should not be inferred from results obtained on simpler models.

3. **Framework reproducibility and explanation method agreement are different checks.**

TensorFlow and PyTorch preserve feature ordering closely, but reimplementing a model in another framework answers a different question from comparing distinct explanation methods. Both framework information and explanation configuration should therefore be documented.

4. **Fairness assessment should not rely on attribution alone.**

Demographic attribution can identify variables that warrant scrutiny, but it does not establish group level disparity. Attribution should therefore be considered alongside formal fairness measures and appropriate uncertainty estimates.

These findings position explanation as one component of a wider model governance process rather than as an automatically authoritative account of model behaviour (Barredo Arrieta et al. 2020).

5.8 Implications for Model Governance

Explanation generation should be treated as part of the analytical pipeline rather than as a purely presentational stage. The explanation method, background or baseline, aggregation procedure, software environment and random seed settings can all affect the resulting output and should be documented alongside conventional model development information. This improves the reproducibility of the explanation as well as that of the predictive model.

For validators and internal auditors, the results support explicit testing of explanation stability. An independent method with a different theoretical basis may reveal uncertainty that simply reproducing the same explanation procedure does not. Agreement between explanations can therefore provide useful evidence, while disagreement identifies areas requiring further investigation.

Framework reproducibility should also be considered separately from explanation method agreement. The German results show that similar feature ordering can coexist with differences in overall attribution magnitude and predictive behaviour. Reproducing a ranking across frameworks therefore does not remove the need to examine the numerical behaviour of the resulting explanations.

For regulated firms, explanation reliability is relevant to the quality of information used around consequential decisions. The FCA Consumer Duty emphasises communications that support consumer understanding (Financial Conduct Authority 2022), while UK data protection legislation retains safeguards around significant automated decision making (United Kingdom 2025). The EU Artificial Intelligence Act provides a separate regulatory example through its treatment of specified creditworthiness systems as high risk (European Union 2024).

The study does not provide a legal compliance test. Its relevance is to governance: where post hoc explanations are used as evidence about model behaviour, uncertainty created by the choice of explanation method or implementation should be visible rather than presenting one explanation as definitive.

5.9 Limitations

Several limitations affect the interpretation and generalisability of the findings.

First, the Taiwan and German datasets are established public benchmarks but do not represent a current United Kingdom lending population. Their demographic results should therefore be interpreted as properties of the evaluated models and datasets rather than estimates of contemporary UK lending disparities.

Second, only two datasets were examined. They differ in sample size, feature structure, categorical representation, target distribution and population. These differences cannot be separated using the present design, and model nonlinearity was not measured directly. The proposed relationship between data availability, model complexity and explanation behaviour is therefore a hypothesis rather than a causal finding.

Third, complete reruns demonstrated some TensorFlow variability despite nominally identical seeds, while LIME introduces additional stochasticity through perturbation sampling. The designated reporting run strategy ensures internal consistency but does not remove this underlying run to run uncertainty.

Fourth, the RQ2 comparison uses ten seeds. Combined with Bonferroni correction, this limits power to detect smaller framework differences. Failure to identify significance therefore cannot be interpreted as equivalence.

Fifth, the model families are limited to logistic regression, XGBoost and one feed forward neural network architecture. Different architectures, regularisation strategies or model families may produce different explanation properties.

The explanation analysis is also limited by its chosen implementations. Different SHAP explainers were required for different model families, with linear, tree based and gradient based approaches used according to model type. Consequently, differences in SHAP behaviour across model families cannot be attributed solely to differences in the predictive models. Gradient based neural network SHAP also shares information with Integrated Gradients, while LIME results depend on perturbation and locality settings. Integrated Gradients used a fixed reference baseline in the transformed feature space. For one hot encoded categorical variables, the straight line path between the baseline and an observation can pass through fractional encoded states that do not correspond to observed categorical combinations, which may affect the resulting attributions. Global mean absolute attribution additionally summarises behaviour across observations and may conceal substantial disagreement for individual applicants.

Finally, the fairness analysis examines only sex and age using the four fifths screening criterion. This captures group level outcome differences but does not identify their cause. The German sex comparison additionally relies on the legacy derived proxy grouping described in Section 3.2.2, because sex cannot be uniquely reconstructed from the corrected Statlog personal status coding (Grömping 2019). It should therefore not be interpreted as a definitive estimate of sex based disparity. Because the statistic is defined as the minimum observed favourable outcome rate divided by the maximum, it also has a downward finite sample tendency. Sampling variation alone can reduce the ratio below one even when the underlying group rates are equal, particularly where some demographic groups contain relatively few observations. Sparse subgroup representation, including the 37 German foreign worker observations, further limits the demographic comparisons. Bootstrap confidence intervals quantify uncertainty but cannot replace information absent from the data.

German model performance estimates also show substantial fold to fold variability, which should be considered when interpreting the predictive comparison.

5.10 Future Work

The most important extension would be to repeat the complete pipeline across a larger collection of credit datasets with different sample sizes, feature structures and demographic compositions. This would test whether the cross dataset patterns identified here recur elsewhere.

A stronger test of the role of data volume would repeatedly subsample a single large dataset while keeping its features, target and population constant. Model complexity could similarly be varied experimentally, with nonlinearity or local decision surface complexity measured directly.

The explanation comparison could be extended using a nongradient SHAP approach to determine whether strong SHAP and Integrated Gradients agreement persists without their shared use of gradient information. LIME sensitivity analysis across perturbation size, locality settings and random seeds would also help identify how much disagreement is configuration dependent.

Future work should additionally examine local explanation agreement. Global

rankings may be similar even where methods identify different leading factors for individual applicants.

For RQ2, more repeated seeds and formal equivalence testing would provide stronger evidence about whether cross framework differences are practically negligible. TensorFlow experiments with deterministic operations explicitly enabled would also help isolate execution variability.

Fairness analysis could be extended using demographic parity differences, group specific error rates, equality of opportunity measures and calibration. Larger demographic samples would reduce the uncertainty observed in smaller groups.

Finally, counterfactual explanations and actionable recourse could complement attribution analysis by examining whether applicants from different groups have comparable realistic opportunities to change an adverse outcome.

5.11 Summary

The results support a conditional view of post hoc explanation reliability. RQ1 shows significant overall concordance but substantial variation between explanation methods, particularly for LIME on the Taiwan neural networks.

RQ2 shows high TensorFlow and PyTorch reproducibility in feature ordering but not complete numerical invariance. RQ3 shows that demographic attribution and group level outcome disparity provide complementary rather than equivalent evidence.

The cross dataset findings are consistent with data availability and model complexity contributing to these patterns, but the two-dataset design does not establish either factor as causal or universally generalisable.

The principal contribution is therefore a reproducible comparative framework and a set of testable empirical observations. Post hoc explanations should be treated as analytical outputs that require validation rather than assumed to provide inherently stable or authoritative descriptions of model behaviour.

Chapter 6

Conclusion

6.1 Summary of Findings

This study examined the reliability of three widely used post hoc explanation methods in credit risk modelling. Four predictive models were evaluated across the Taiwan Default of Credit Card Clients and German Statlog Credit datasets, with three aspects of explanation reliability considered: agreement between explanation methods, reproducibility across TensorFlow and PyTorch implementations, and the relationship between feature attribution and demographic outcome disparities.

Three main findings emerged.

First, explanation method agreement is conditional rather than guaranteed. Overall Kendall's W was statistically significant for all four neural network analyses, but pairwise agreement varied considerably. SHAP and Integrated Gradients remained strongly correlated, with Spearman coefficients ranging from 0.890 to 0.952, whereas comparisons involving LIME were substantially weaker on the Taiwan neural networks. RQ1 therefore shows that explanation agreement should be measured within the model and data setting rather than assumed from the explanation method selected.

Second, TensorFlow and PyTorch produced highly similar feature rankings, with framework level Spearman correlations of 0.957 on Taiwan and 0.962 on German. No Taiwan features showed statistically detectable differences after Bonferroni correction, whereas two of the 20 German features did. Attribution magnitudes nevertheless differed, particularly on German. RQ2 therefore supports high reproducibility of feature ordering rather than formal equivalence between independently trained implementations.

Third, demographic attribution and outcome disparity provided complementary rather than interchangeable evidence. Taiwan logistic regression produced an age

ratio of 0.680 with a bootstrap confidence interval entirely below 0.8. On German, all four age ratios and their complete confidence intervals remained below the four fifths screening threshold. However, direct demographic attribution did not consistently correspond with outcome disparity. RQ3 therefore demonstrates that explanation alone cannot replace formal fairness assessment.

A further contribution of the study is the use of formal statistical procedures to evaluate each of these questions rather than relying on visual comparison alone. Kendall’s concordance and pairwise rank correlations measure explanation agreement, paired Wilcoxon procedures with Bonferroni correction assess framework differences, and bootstrap confidence intervals quantify uncertainty in demographic outcome ratios.

Across the two datasets, the results are consistent with data availability and fitted model complexity contributing to differences in explanation behaviour. This remains a testable interpretation rather than a causal conclusion because only two datasets were examined and their characteristics differ in several respects. The findings therefore support a conditional account of explanation reliability rather than a universal claim about any individual explanation method or framework.

6.2 Limitations

The findings should be interpreted within several limitations. The Taiwan and German datasets are established public benchmarks but do not represent a current United Kingdom lending population. Their demographic results therefore describe the evaluated data and models rather than contemporary UK lending practices.

Only two datasets and a limited set of model families were examined. Dataset size, feature composition, population and class distribution vary simultaneously, while model nonlinearity was not measured directly. The proposed relationship between data conditions, model complexity and explanation behaviour cannot therefore be interpreted causally.

The framework comparison also used only ten random seeds. The absence of statistically significant feature differences does not demonstrate equivalence, particularly after conservative Bonferroni correction. TensorFlow results showed some run to run variation, while LIME also contains stochastic variation through perturbation

sampling.

Finally, the analysis focuses mainly on global feature rankings and a four fifths group level screening measure. Global agreement can conceal local disagreement between individual explanations, while the four fifths ratio cannot determine the cause or legal significance of an observed demographic disparity. These limitations restrict the scope of the numerical conclusions but do not prevent the analytical procedure from being applied to broader settings.

6.3 Practical Implications

The findings support three practical conclusions.

First, post hoc explanation methods should be validated rather than treated as automatically reliable. The Taiwan neural network results demonstrate that method choice can materially alter the resulting feature ranking.

Second, framework reproducibility and explanation method agreement should be considered separately. TensorFlow and PyTorch produced very similar feature ordering, but this does not remove uncertainty introduced by the explanation method itself. Software environment, framework version, random seeds and explanation configuration should therefore remain part of the reproducibility record.

Third, explanation should be used alongside formal fairness analysis. Demographic feature importance can identify variables requiring further scrutiny, but it does not establish whether favourable outcomes differ systematically between demographic groups. Model governance should therefore treat attribution, reproducibility and fairness assessment as related but distinct checks.

6.4 Closing Remarks

Post hoc explanation methods can provide useful evidence about credit risk models, but their reliability is not automatic. SHAP and Integrated Gradients showed strong agreement in the evaluated models, while LIME was more dependent on the model and data setting. TensorFlow and PyTorch preserved feature rankings closely, but numerical attribution and predictive differences remained. Demographic

attribution and group level outcome measures similarly provided related but distinct evidence.

The principal contribution of this study is therefore a reproducible framework for evaluating explanation method agreement, implementation reproducibility and demographic outcome disparity, together with a conditional account of explanation reliability. The numerical findings belong to the two datasets examined, but the analytical procedure can be applied to additional datasets to determine whether the observed patterns persist under different conditions. For consequential financial decisions, explanations should be treated as analytical evidence rather than simply as outputs of the model development process. Their agreement, reproducibility, uncertainty and relationship to fairness should be assessed before they are relied upon as authoritative descriptions of model behaviour.

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Appendix A

Reproducibility and Supporting Material

A.1 Overview

This appendix provides the supporting evidence required to reproduce and inspect the experimental analysis described in Chapter 3. It identifies the principal analysis components, records the order in which the analytical stages were carried out, summarises the software environment and explanation configuration, and documents how reporting consistency was maintained.

The appendix is intended to support transparency and reproducibility rather than to repeat the principal statistical findings presented in Chapter 4. The numerical results reported in the dissertation are taken from the designated complete reporting run described in Section A.9.

A.2 Reproducibility Repository

The modelling, explanation and statistical analysis code developed for this study is maintained in the following repository:

<https://github.com/GDavis-01/Explainable-AI-for-Credit-Risk-Assessment>

The version corresponding to the submitted dissertation is identified by Git commit `b5b42fb`.

The repository is used to retain the Python analysis notebooks, R statistical analysis code, intermediate CSV files and supporting documentation associated with the experimental workflow.

The dissertation describes the methodological decisions and principal findings, while the repository provides the executable material required to inspect, reproduce and

extend the analysis. The exact organisation and filenames in the repository are documented alongside the final code so that the analysis components described in Section A.3 can be matched directly to their implementation.

A.3 Analysis Components and Their Purpose

The analytical workflow was separated into distinct components so that the two datasets could be processed independently and the formal statistical analysis could be separated from model fitting and explanation generation.

Table A.1 identifies the principal analysis components, the environment used for each stage and their role within the reproducible workflow.

Table A.1: Principal analysis components and their role in the experimental workflow.

Analysis component	Environment	Purpose
Taiwan analysis	Python / Google Colab	Data acquisition and preprocessing for the Taiwan dataset; logistic regression, XGBoost, TensorFlow and PyTorch model fitting; predictive evaluation; SHAP, LIME and Integrated Gradients attribution; fairness inputs; and export of intermediate results for statistical analysis.
German analysis	Python / Google Colab	Data acquisition and preprocessing for the German dataset; fitting of the four predictive models; predictive evaluation; SHAP, LIME and Integrated Gradients attribution; framework comparison outputs; fairness inputs; and export of intermediate results.
Supporting experimental analysis	Python / Google Colab	Preparation of additional outputs required for the comparative experimental workflow, including material used for selected analytical figures and cross dataset comparisons.
Formal statistical analysis	R / RStudio	Kendall's coefficient of concordance, pairwise Spearman correlations, paired Wilcoxon signed rank tests, Bonferroni adjustment, bootstrap confidence intervals, Friedman tests, Nemenyi post hoc comparisons, and generation of the principal statistical figures.

Separating the Taiwan and German analyses reduces the possibility of variables or intermediate state being transferred unintentionally between datasets. The R analysis is similarly separated from the modelling stage so that the formal statistical procedures operate on explicitly exported results rather than directly on transient notebook objects.

A.4 Reproduction Workflow

The complete analysis was carried out in the following order:

1. The Taiwan dataset was obtained from its original repository and processed in the Taiwan Python analysis.
2. The four Taiwan predictive models were trained and evaluated.
3. SHAP and LIME explanations were generated for the applicable models, together with Integrated Gradients for the two neural networks.
4. Outputs required for explanation concordance, framework reproducibility and fairness analysis were exported from Python.
5. The German dataset was obtained and independently processed in the German Python analysis.
6. The corresponding German models, explanation methods and exported analysis files were generated using the same overall experimental structure.
7. The repeated seed TensorFlow and PyTorch framework comparison was performed and the feature level outputs were exported.
8. The exported results were imported into RStudio.
9. Formal statistical analyses were performed in R, including Kendall's coefficient of concordance, pairwise Spearman correlations, paired Wilcoxon tests, Bonferroni correction, bootstrap confidence intervals, Friedman tests and Nemenyi comparisons.
10. Statistical figures and summary tables were generated from the resulting outputs.
11. One complete internally consistent execution was designated as the primary reporting run used for the numerical tables and analytical figures in the dissertation.

This sequence provides an auditable path from the original datasets through model construction and explanation generation to the formal statistical results reported in Chapter 4.

A.5 Software Environment

Table A.2 records the principal software and library versions used in the designated reporting analysis, providing the environment information required to support reproducibility.

Table A.2: Principal software used in the experimental analysis.

Software or library	Version / environment
Python	Google Colab environment
TensorFlow	2.20.0
PyTorch	2.11.0+cpu
scikit-learn	1.6.1
XGBoost	3.3.0
R	4.4.1
RStudio	Statistical analysis environment
LaTeX	Overleaf

Package versions are included because numerical machine learning results can depend on implementation details that change between software releases. They therefore form part of the reproducibility evidence rather than being treated as incidental implementation information.

A.6 Random Seeds and Repeated Execution

Random seeds were fixed where supported in order to reduce unnecessary stochastic variation. For the principal TensorFlow and PyTorch framework comparison, ten matched seed values were used to provide paired observations for the statistical analysis.

For the R statistical analysis, the random seed was set to 42 before bootstrap resampling so that the reported bootstrap confidence intervals could be reproduced consistently.

Setting a random seed does not necessarily guarantee completely deterministic execution. In particular, TensorFlow operation determinism was not explicitly

enabled during the reported experiment. Separate complete executions therefore produced some variation in TensorFlow derived explanation results even when nominal seed settings were retained.

LIME also relies on stochastic perturbation sampling, meaning that quantities involving LIME may exhibit small differences between otherwise comparable executions.

For this reason, results from different complete runs were not mixed when constructing the main dissertation tables and figures. One complete run was designated as the reporting run, and its numerical outputs were used consistently throughout the principal analysis. Separate executions were retained as reproducibility evidence rather than being combined with the designated reporting results. The distinction between the reporting run and supporting reruns is documented in Section A.9.

A.7 Explanation Configuration

The explanation methods were configured consistently within each model family.

For logistic regression, SHAP was calculated using the linear explainer. XGBoost was analysed using TreeSHAP. The TensorFlow and PyTorch neural networks were analysed using a gradient based SHAP explainer with a fixed background sample of 200 training observations.

LIME was applied to all four model families using classification mode. The principal analysis used 1,000 perturbation samples and continuous feature discretisation was disabled. The explanation configuration was held fixed for the primary comparison rather than tuned separately for individual models.

Integrated Gradients was applied only to the differentiable TensorFlow and PyTorch neural networks. The baseline represented the reference state of the transformed feature space: the training mean for standardised numerical variables and the inactive or reference level for binary and one hot encoded variables.

The PyTorch implementation used Captum with 64 integration steps. The TensorFlow implementation used automatic differentiation and a numerical approximation to the Integrated Gradients path described in Chapter 3.

These settings form part of the reproducibility record because attribution results depend not only on the explanation method selected but also on the configuration used to implement that method.

A.8 Statistical Analysis Procedures

The formal statistical analysis was carried out in R after the relevant Python outputs had been exported.

The principal procedures were:

- Kendall's coefficient of concordance with tie correction for the three method neural network comparisons;
- pairwise Spearman rank correlation for explanation ranking comparisons;
- paired Wilcoxon signed rank tests for TensorFlow and PyTorch feature

attribution comparisons across the ten matched seeds;

- Bonferroni correction across the feature level framework comparisons;
- nonparametric bootstrap confidence intervals based on 2,000 resamples for the four fifths ratios;
- Friedman tests for comparison of model performance across the shared cross validation folds; and
- Nemenyi post hoc comparisons where the Friedman omnibus test was statistically significant.

These procedures correspond to the statistical methods defined in Chapter 3 and provide the formal tests underlying the results reported in Chapter 4.

The R analysis also generated the statistical figures used in the Results chapter, including the concordance heatmap, cross validation AUC comparison, demographic fairness ratio figure and associated comparative plots.

A.9 Reporting Run and Internal Consistency

A designated reporting run was used to prevent results from separate stochastic executions being combined inadvertently.

The values in the principal Results tables, figures and accompanying text originate from the same complete analytical execution unless explicitly identified as reproducibility results from a separate run.

This distinction is particularly relevant to TensorFlow and LIME because some outputs showed movement between complete executions. The purpose of the additional runs was to examine reproducibility rather than to select whichever numerical result was most favourable.

Maintaining a single designated reporting run ensures that the reported feature rankings, concordance statistics, framework comparisons, fairness ratios and associated figures remain internally consistent. The separate complete executions reported in Section 4.4.1 are therefore supporting evidence concerning reproducibility rather than alternative reporting results.

A.10 Supporting Outputs and Audit Trail

The complete analysis produces substantially more intermediate information than can be presented within the main dissertation. This includes full feature level attribution outputs, seed level framework comparisons, bootstrap results, individual cross validation values and intermediate CSV files.

These materials form part of the audit trail for the analysis and are retained alongside the executable code rather than reproduced in full within the dissertation. They allow the numerical inputs to the formal statistical procedures to be inspected independently of the final tables and figures.

The main dissertation therefore presents the evidence required to answer the research questions, while the supporting outputs preserve the additional material needed for independent inspection and reproduction. This separation keeps the main report focused while ensuring that the underlying analytical evidence remains traceable.